Semantic Segmentation for Remote Sensing Image Using the Multigranularity Object-Based Markov Random Field With Blinking Coefficient

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***Abstract*— Semantic segmentation is one of the most important tasks in remote sensing. In the semantic segmentation of remote sensing images, some regions are repeatedly transformed between multiclasses, which affects the convergence speed and segmentation accuracy. This is because the increased spatial resolution makes the spectral distribution of geographic targets differ from the overall category distribution. The Markov random field (MRF) model is widely used for semantic segmentation of remote sensing images because of its outstanding spatial description ability. Some scholars have made improvements on MRF models to extract more information or enhance semantic inference. However, these improvements fail to capture the correlation between the multigranularity layers and the historical information. In this article, we propose a new MRF-based model that adopts multigranularity layers to realize the multigranularity correlation representation of targets and the spatial–temporal inference of segmentation labels. First, the algorithm constructs a multigrained layer structure based on remote sensing images to enhance feature extraction for targets of different sizes in images. Second, for the multilayer feature field, a cross-layer Gauss–Markov model is constructed based on intra–inter-layer feature correlation constraints. Then, for the multigranularity layer label field, a self-renewing pairwise spatial–temporal potential function with blinking coefficients is constructed based on the newly defined cross-layer augmented neighborhood system, which can accelerate the convergence of segmentation by using the history information and spatial neighborhood information. The proposed method is tested on texture images, SPOT-5, and Gaofen-2 images. Experiments show that the proposed method has a better performance compared to other state-of-the-art MRF-based methods.**

***Index Terms*— Markov random field (MRF), multigranularity, remote sensing image, semantic segmentation.**

Manuscript received 12 June 2023; revised 3 July 2023; accepted 23 July 2023. Date of publication 3 August 2023; date of current version 16 August 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 61976080, Grant 52177109, and Grant 51707135; in part by the Hubei Province Key Research and Development (R&D) Program under Project 2020BAB109; and in part by the Natural Science Project of the Zhengzhou Science and Technology Bureau under Grant 22ZZRDZX43. *(Corresponding author: Hongtai Yao.)*

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Digital Object Identifier 10.1109/TGRS.2023.3301494

I. Introduction

SEMANTIC segmentation for remote sensing images is a very important task; it can provide basic data for many geographic information analysis, such as coastline prediction [1], [2], agricultural land statistics [3], [4], and urban greening analysis [5], [6]. The more accurate the semantic segmentation results of remote sensing images are, the conclusions obtained from the following geographic information analysis can be more in line with objective reality and the law of future development.

At present, researchers have proposed a variety of methods about semantic segmentation for remote sensing images, including superpixel [7], [8], [9], level-sets [10], [11], [12], [13], active contours [14], [15], [16], [17], Bayesian networks [18], [19], [20], [21], methods based on random fields [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], and methods based on deep learning [9], [33], [34], [35], [36], [37], [38], [39], [40]. Deep-learning-based methods usually use end-to-end learning, where the feature extraction is influenced by a back-propagation mechanism, so that the parameters of the feature extractor and classifier are jointly optimized to achieve overall optimality. There are three main deep learning semantic segmentation algorithms that are commonly used, which are the pyramid structure-based algorithm [41], [42], the encoder–decoder-based algorithm [41], [43], and the transformer-based algorithm [44]. These network structures can extract deeper abstract features of images to better explore the feature representation of different targets, and they can continuously learn the contextual associations between different feature blocks so that they have higher segmentation accuracy. However, the segmentation accuracy of these methods is largely related to the quantity and quality of training data, and it requires that the test data obey the same distribution as the training data. The random field-based method gives a way to semantic interpretation from the view of statistic: not only the spectral value information of the images is used, but also the

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spatial structure information is captured by constructing a neighborhood system. Methods based on random field can be roughly divided into two types: 1) the conditional random field (CRF)-based methods [8], [25], [27], [31], [32] and 2) the Markov random field (MRF)-based methods [22], [29], [30]. These two kinds of methods are both based on probability and statistics. While the CRF-based method is a type of discriminative model, the MRF-based method is a type of generative model. In this article, our research belongs to the latter. In particular, we propose a multigranularity object-based MRF with blinking coefficients (MGOMRF-BC), which can mine the different granularities feature representations of the same target in remote sensing images and can jump out of the local loop when updating the label field.

With the increasing resolution of remote sensing images, the same geographic target can be represented by more pixels, and the difference in spectral values between geographic targets becomes larger, which poses a huge challenge for semantic segmentation. This is because the difference in spectral values between the same classes of targets becomes larger, while there is spectral overlap between different classes of targets, resulting in fragmentation misclassification in the segmentation results.

Whether it is a deep learning algorithm or a traditional machine learning algorithm, some pixels or regions will be repeatedly transformed among several categories, which will eventually cause the misclassification of regions. This is because, with the development of sensor technology, the resolution of images returned by remote sensing satellites has become higher, and the description of geographic targets has become clearer. Compared with the low-spatial resolution remote sensing images in the past, in the current high-spatial resolution remote sensing images, the difference in the spectral values of some different subcategories that belong to the same category is more obvious, and some subcategories that do not belong to the same category have similar spectral values. These regions are repeatedly transformed between several semantic categories when the segmentation labels are updated. To solve this problem, many solutions have been proposed; they can be broadly divided into two categories: 1) extracting deep features that can distinguish each semantic category and 2) using spatial and semantic information of small-sized regions of remote sensing images and their surrounding regions for semantic inference. In particular, for the former, there are some deep learning algorithms that focus on extracting multiscale information in images in order to get feature of the same target in different sizes, for example: 1) Zhang et al. [41] construct the atrous spatial pyramid pooling (ASPP) block [45] based on depthwise separable convolution (SASPP) and squeeze-and-excitation (SE) channel attention block to enhance extraction of multiscale feature information; 2) Xiao et al. [44] proposed Enhancing Multiscale Representations with Transformer (EMRT) to enhance multiscale representation learning; and 3) Luo et al. [42] propose a multistructure unified discriminative embedding (MUDE) method, which can integrate the complementarity between different features to achieve the complementarity of different features. There are also some MRF-based algorithms that are combined with multiscale structures to obtain deep features in images through multiresolution information, for example: 1) Hong et al. [23] proposed the hierarchically multiscale object-oriented MRF (HMSOMRF) model, which can use the spectral, textural, and shape features of each node as the feature vector and 2) Zheng et al. [46] proposed multigranularity multiclass-layer MRF (MGMCL-MRF) model, which builds a hybrid probability graph on both pixel and object granularities to incorporate various granularity features and multilayer semantic information. As for the latter, the idea of inference with the help of spatial information and semantic information around the region has also been investigated by many scholars, such as: 1) Liu et al. [47] proposed a new unsupervised segmentation method HMRF-FCM, which introduces region information into the label field of MRF and 2) Zheng et al. [30] proposed Object-Based MRF Model With Auxiliary Label Fields (OMRF-A), which establishes multilayer labels random fields to assist segmentation tasks and obtains more accurate results.

These two ideas still have some shortcomings in solving the misclassification problem of small-area targets: 1) in extracting deep features, the correlation between these features is not reasonably modeled, and the deep learning algorithm is a direct splicing, while the MRF algorithm that combines multiscale is only an independent modeling of the feature field and 2) when semantic inference is used, the neighborhood information of a single scale is considered, and the repeated transformation of small regions between several semantic categories in the updating process cannot be prevented. This is because the spectral features of these regions are in the overlapping regions of the spectral distributions among several semantic categories, and it is difficult to use the segmentation labels of the surrounding adjacent regions as a reference for the semantic labels of these small regions.

In order to better feature extraction and model the small target regions of remote sensing images, to be able to make full use of spatial information in the updating process while accelerating the convergence of segmentation labels, inspired by the idea of multigranularity and semantic information inference, we propose an MGOMRF-BC in this article. The algorithm first constructs a multigranularity structure. Second, intra–inter-layer correlation object-based Gauss–Markov (G-M) models, which across granularity layers, are constructed. Finally, the blinking coefficients are defined and employed in the label field potential function when the label field is cyclically updated in each layer; it establishes the spatial–temporal relationship between each granularity layer itself and its adjacent granularity layers, accelerates the convergence speed of semantic segmentation, and improves the regional consistency of semantic segmentation results in images.

The main contributions mentioned in this article are given as follows.

1) The multigranularity layers are used to form different sizes of oversegmented region results; this can effectively reduce misclassification caused by dividing adjacent pixel points of different classes into the same

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oversegmented region in the oversegmentation stage. We treat these oversegmented regions as objects.

2) By establishing a multigranularity layer structure, different granularities' information of the same geographic target is extracted, and the object-level G-M model based on the correlation assumption is used for feature modeling, which can fully consider the interactions of intra–inter layers between granularity features contained in the adjacent layer.
3) In the process of modeling the label field of each granularity layer, not only the neighbors of this layer but also the corresponding neighbors of adjacent granularity layers are considered into the modeling, and the blinking coefficient is added to the potential energy function to accelerate the convergence of the semantic segmentation results.

The other parts of this article are organized as follows. Section II briefly explains the improvements based on MRF model in the past few years. In Section III, we will introduce the MGOMRF-BC algorithm in detail. In Section IV, we discuss the experimental results of MGOMRF-BC on various remote sensing images. Finally, Section V concludes this article.

II. Related Work

MRF-based algorithms are widely used in the task of semantic segmentation for remote sensing images. It establishes two random fields on an image, namely, the label field and the feature field. The label field is used to describe the spatial adjacency information of each node in the image and record the segmentation results, while the feature field is used to describe the features of each node. When the image is given, the realization of the feature field is unique. Assuming that the label field has Markov property and a prior segmentation is given, the segmentation problem is transformed into a problem of maximizing the posterior segmentation probability. The probabilities of two subrandom fields are modeled; combining Bayes rule and maximum a posteriori (MAP) criterion, the final segmentation result can be obtained. The MRF-based algorithms can not only describe the spectral feature distribution of an image but also capture the spatial distribution relationships presented among the various feature classes in an image by introducing Markov properties. At present, most of MRF-based algorithms for semantic segmentation of remote sensing images are either concerned with the multigranularity representation of the image for better description of the feature targets or with the probabilistic construction of the label fields in the modeling so that the Markov property can better play the role of "clustering things together." The relevant work in these two areas is summarized in the following.

A. MRF Model With Multigranularity

The classical MRF algorithm uses pixel points as the nodes to be processed and defines the feature field and label field on the pixel grid, respectively. Among them, the feature field uses a likelihood function to describe the probability of distribution of features for each pixel point given a segmentation realization, and the label field uses a Gibbs random field to construct the probability of spatial distribution of segmentation labels for each pixel point. With the increasing spatial resolution of remote sensing images, the complex texture shapes and large spectral differences make the pixel-level MRF method unable to accurately describe the features of the targets, as well as the likelihood function. In order to make better use of the detailed information brought by high spatial resolution, object-based image analysis (OBIA) [48] has been introduced into the semantic segmentation of remote sensing images, and a series of geographic OBIA (GEOBIA) [49] algorithms have emerged. These algorithms can not only effectively reduce the number of nodes to be processed but also effectively smooth out the differences in spectral values of targets and mitigate the misclassification phenomenon. However, it is still far from solving the semantic segmentation of remote sensing images.

The multigranularity idea was initially identified at the pixel level technology as a multiscale structure. A given remote sensing image can be constructed with a multiscale structure to mine the scale information contained in the image, and the multiscale representation can visually reduce the image size to achieve coarse-granularity representation. Noda et al. [50] employed the wavelet decomposition to get multiresolution region image and used doubly stochastic MRFs to capture intrascale and interscale dependencies over multiresolution region image. Liu et al. [51] built a deep parsing network (DPN) to model unary terms, and additional layers are devised to approximate the mean field (MF) algorithm. After GEOBIA was proposed, the multigranularity technique is more widely used. Xia et al. [52] obtained the region adjacency graph (RAG) with the watershed algorithm and constructed an object-based MRF by defining a region-based likelihood function and prior probability on the RAG. Zheng et al. [53] extends the multiresolution from the pixel level to the object level and incorporated the multigranularity information and the MRF. In [54] and [55], different hybrid MRF models are built to capture the multigranularity information, and the corresponding hybrid label field realizes the synergy between the multigranularity information and the hybrid label interactions. Yao et al. [56] employed multigranularity and multisemantic ideas in the multiscale structure to construct a new pyramid structure to capture feature correlations and spatial–semantic role relationships at different resolutions and different granularities.

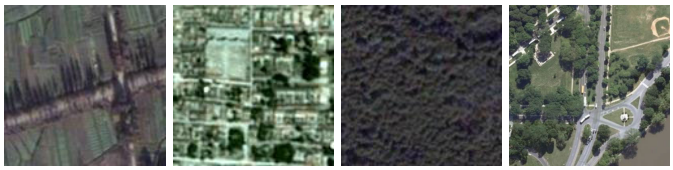
These MRF-based models with multigranularity can effectively mine the features with different granularities in the images and improve the segmentation accuracy of remote sensing images. However, most of them adopt the independence assumption in modeling the feature fields of a single granularity layer and then add interscale action functions, which ignores the interactions of the corresponding location features between adjacent granularity layers.

B. MRF Model With Semantic Inference

In high spatial resolution remote sensing images, many subcategories are included, and there are overlapping regions of spectral values between each subcategory.

As shown in Fig. 1, a tree can naturally be classified as a tree when it exists independently. However, if it is placed on

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(a) (b) (c) (d)

Fig. 1. Example for semantic inference. (a) Trees by the path in the field, (b) trees in the city, (c) trees in the forest, and (d) trees in the park.

both sides of the village street, it belongs to the road class; if it is placed inside the town park, it belongs to the park class; if it is placed in the wild forest, it belongs to the forest class; and if it is placed in the green belt on both sides of the road in the city, it belongs to the town green class. This is because, for a remote sensing image, it contains a large number of targets, and these targets are inextricably linked with each other, and different semantics can be deciphered from different perspectives.

For this reason, semantic inference has been added by many people in the process of modeling label fields for MRF-based to avoid misclassification due to interclass similarity or intraclass differences of spectral features. Semantic inference means how to use the semantic information of surrounding neighboring nodes to infer the semantic category affiliation of the current node. In MRF-based algorithms, the most basic semantic inference is based on the Markov property, which can be represent as "clustering things together": the semantic category of the current node is largely influenced by the semantic categories of its neighboring nodes, as shown in Fig. 2(a) and (b); when most of the surrounding nodes belong to a certain semantic category, the current node will most likely belong to that semantic category. There may also be cases where the surrounding nodes belong to different categories, just like Fig. 2(c) and (d), and it is necessary to combine the probability of feature distribution of the current node to make a comprehensive judgment. However, for high-resolution remote sensing images, there will be some small target regions with close spectral values and adjacent spatial locations but not belonging to the same category, which will be misclassified under the effect of Markov property, and then, some scholars have proposed more complex semantic inference mechanisms to solve this misclassification phenomenon, as shown in Fig. 2(e) and (f): the concept of auxiliary label field is introduced in the label field modeling process, and by setting the number of different semantic categories from the standard semantic categories in the auxiliary label field, the results of both are combined to provide reference semantic information for the standard semantic categories of the current node.

Zhang et al. [57] identify and fuse multiple segmentation fields with the help of segmentation tree models to achieve semantic segmentation optimization for specific targets. Tuia et al. [58] improve the energy function in the random field model, introduce geographic information into the modeling of prior probabilities, and define the CRF on a bipartite graph structure to realize the semantic transfer role between subgraphs. Zheng and Wang [29] introduced both region size and edge information to construct the anisotropic potential energy function. In [46], the multigranularity multiclass-layer MRF model was proposed to capture hierarchical semantic information hidden in different granularityies layers. Yao et al. [59] achieved semantic inference on some small target regions by constructing partition maps and performing global–local alternate updates on the images.

These MRF methods with semantic inference usually define auxiliary segmentation label fields to achieve a deterministic division of categories for subcategories in each category. However, it ignores the variability and iterative transformation of semantic segmentation results at different granularities, which affects the segmentation accuracy. In the proposed MGOMRF-BC algorithm, not only the feature correlation of the corresponding positions of the multigranularity layers is considered but also the blinking variation of the segmentation results of the corresponding positions of the multigranularity layers is incorporated into the label field modeling to achieve semantic inference within the multigranularity information intra–inter layer. Both feature correlations and semantic blinking coefficients contained in the multigranularity layers can be learned in the MGOMRF-BC model.

III. Proposed Method

A. MRF Method

MRF is a common image segmentation algorithm based on the probabilistic graph model. When an image I of size $M \times N$ is given, the position index set of all pixels in the image is denoted as $S = \{(m, n)|1 \le m \le M, 1 \le n \le N\}$. We can get the graph model $G = (V, E)$, where $V = \{V_i|1 \le i \le n\}$ and $E = [e_{i,j}]_{n\times n}$. V represents the node set of the image I, and E represents the adjacency relationship between the nodes. For the pixel-based method, v_i is the position index of the pixel, while, for the object-based method, v_i represents the set of position index of pixels contained in the oversegmented region.

The MRF algorithm can be defined on the graph model G; according to the node set V and topological relationship E, two random fields can be established, respectively. One used to describe the features of each node called a feature field, denoted as $Y = \{Y_i|1 \le i \le n\}$; another used to record segmentation labels of each node called label field, denoted as $X = \{X_i|1 \le i \le n\}$. Y_i is the feature vector of node V_i; when image I is given, Y_i is fixed; and X_i is a random variable, which represents the segmentation result of node V_i. We segment the image into K categories, $X_i \in \Lambda$, and $\Lambda = \{1, 2, \ldots, K\}$. We assume that the labels field X is a random field with Markov property, namely,

$$P\left(X_i \middle| X_{V/V_i}\right) = P\left(X_i \middle| X_{N_i}\right) \tag{1}$$

where X_{V/V_i} represents the segmentation label of all nodes except the node V_i, N_i represents the neighborhood node of node V_i, $N_i = \{j|e_{i,j} \neq 0\}$, and X_{N_i} represents the segmentation labels of the neighboring nodes of the node V_i.

The problem, semantic segmentation for remote sensing images, can be transformed into the following: under the premise that the realization of the feature field Y is y, how to

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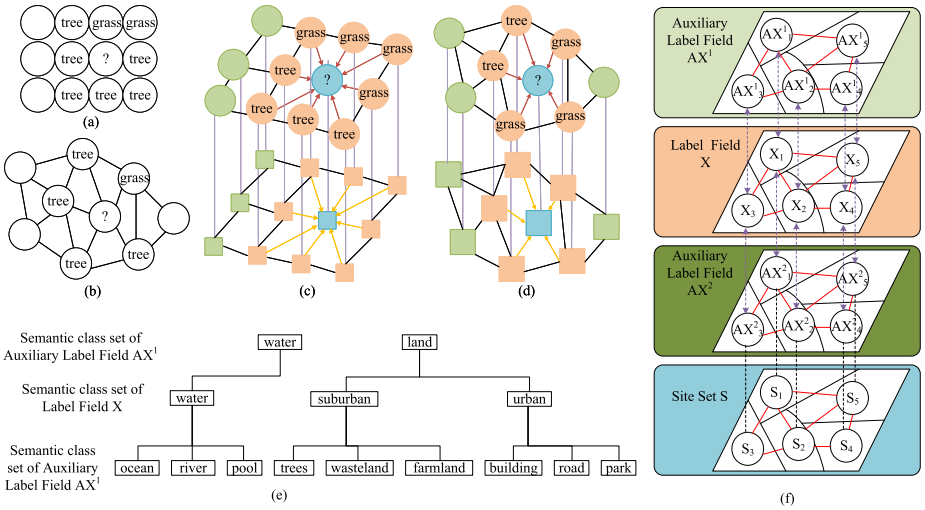


Fig. 2. Several forms of semantic inference. (a) and (b) Pixel and object level neighborhood semantic inference (the semantic categories of surrounding neighborhoods are basically consistent). (c) and (d) Pixel and object level neighborhood semantic inference (the semantic categories of surrounding neighborhoods are different). (e) Different semantic depths. (f) Semantic inference with auxiliary label fields.

maximize the probability of the realization of the label field X as x, namely,

$$x = \arg\max_{x} P(X = x|Y = y). \tag{2}$$

According to the Bayesian formula, the above formula can be transformed into

$$\begin{aligned} x &= \arg\max_{x} P(X = x|Y = y) \\ &= \arg\max_{x} P(Y = y|X = x) \cdot P(X = x). \end{aligned} \tag{3}$$

In the above formula, $P(X = x)$ is the prior probability when the label field $X = x$; $P(X = x)$ obeys Gibbs distribution, i.e.,

$$P(X = x) = \frac{1}{Z} \cdot \exp(-U(x)) \tag{4}$$

where Z is a normalization constant and $U(x)$ is the sum of the potential energy functions of all cliques.

$P(Y = y|X = x)$ represents the likelihood of the feature field $Y = y$ when the label field $X = x$. We assume that all nodes are independent after the segmentation labels are given, and the likelihood probability of each node obeys the mixed-Gaussian distribution, i.e.,

$$\begin{aligned} &P(Y_i = y_i|X_i = k) \\ &= \frac{1}{(2\pi)^{D/2} \cdot |\Sigma_k|^{1/2}} \\ &\quad \cdot \exp\left[-\frac{1}{2}(y_i - \mu_k)^T \cdot (\Sigma_k)^{-1} \cdot (y_i - \mu_k)\right] \end{aligned} \tag{5}$$

where D is the dimension of y_i, and μ_k and Σ_k are the parameter of mixed-Gaussian distribution for the kth category.

B. Framework of the Proposed MGOMRF-BC

In the MRF-based methods, there is a common problem that needs to be solved in the segmentation update iteration process, and there are always some nodes that change among several categories. This is because, at each iteration, the features of these nodes are near the feature averages of several categories, and it is not easy to fix them into their own categories. This phenomenon will slow down convergence speed and cause misclassification in the final segmentation result. In order to fully exploit the information at different granularities and accelerate the convergence of segmentation results using semantic inference in segmentation label updates, as shown in Fig. 3, the framework of the MGOMRF-BC is elaborated on in this section.

In the MGOMRF-BC model, based on a given remote sensing image, different granularities can be considered simultaneously. It takes the oversegmented region as the object to be processed. For a given remotely sensed image I, it is divided into K semantic categories. As shown in Fig. 3, using different settings of the oversegmentation algorithm, L layers with different granularities can be obtained. The coarsest granularity layer is the Lth layer, and the pixel-level granularity is the zeroth layer. The cite index set of the lth level is denoted as S^l. Then, a hybrid probability map $G = \{G^l | 0 \le l \le L\}$, $G^l = (V^l, E^l)$, can be defined on the set of these granularity region nodes site set $S = \{S^l | 0 \le l \le L\}$, where V^l is all nodes in the lth granularity layer and E^l is the matrix recording the adjacency of all nodes in the lth granularity layer. On the multigranularity probability graph structure G, the label fields $X^l = \{X_s^l | s \in S^l\}$ and the feature fields $Y^l = \{Y_s^l | s \in S^l\}$ can be constructed. The label field X^l is a random field with Markovness, X_s^l is a random variable that takes values in $\Lambda = \{1, 2, \ldots, K\}$, and x^l is a realization of X^l. The feature field Y^l has a unique realization y^l after the image I is given, and its probability function is a measure of the likelihood that the realization of the feature field Y^l is y^l when the label field $X^l = x^l$.

Since then, the original problem of optimally solving the segmentation results of a single layer in the traditional MRF algorithm is transformed into the problem of jointly optimally solving the segmentation results of multigranularity layers in MGOMRF-BC, as shown in the following equation:

$$\begin{aligned} &\hat{x}^0, \hat{x}^1, \ldots, \hat{x}^L \\ &\quad = \arg\max_{x^0, x^1, \ldots, x^L} P\left(x^0, x^1, \ldots, x^L \middle| y^0, y^1, \ldots, y^L\right) \end{aligned}$$

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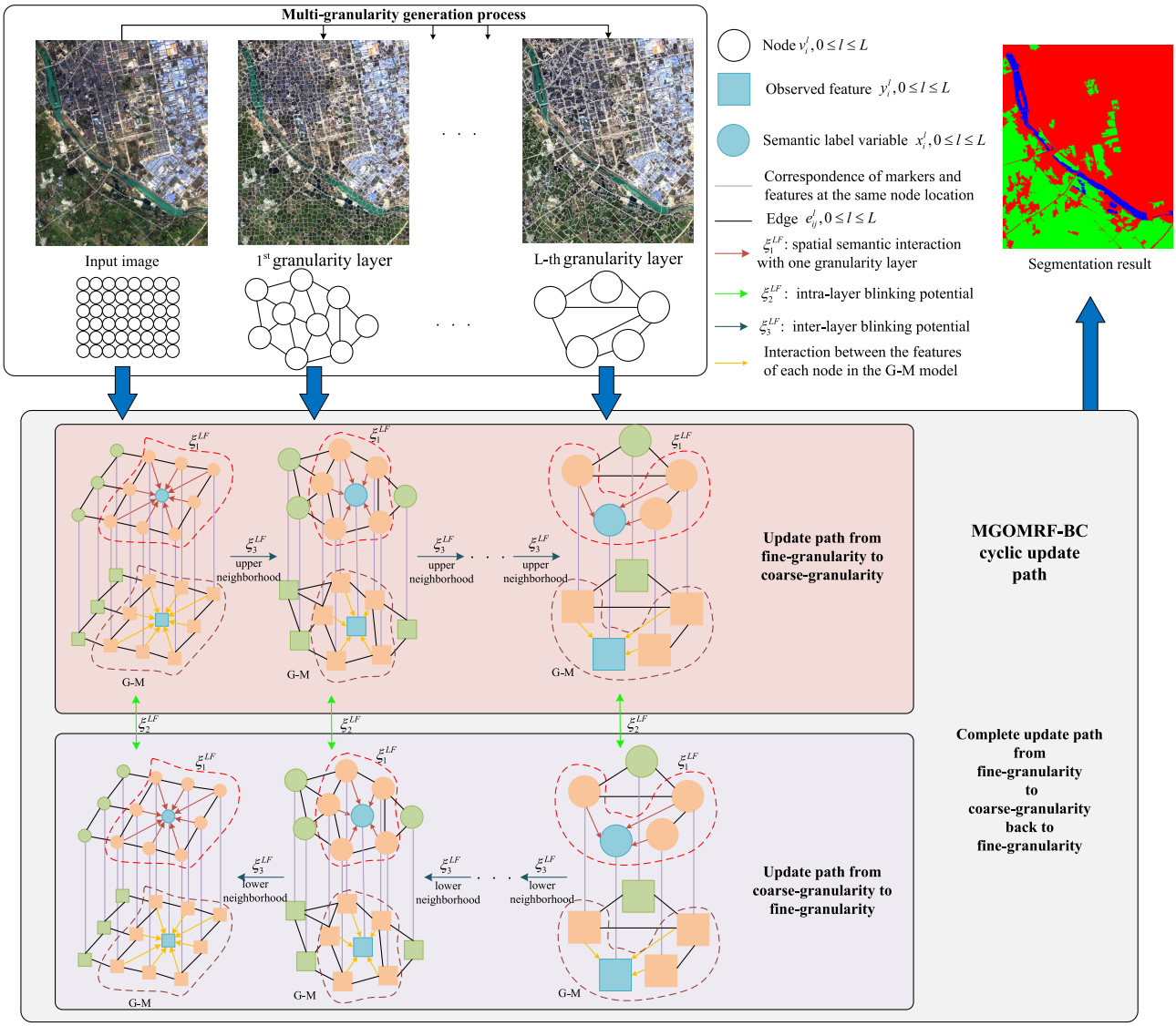


Fig. 3. Framework of the MGOMRF-BC model.

$$\begin{aligned} &\propto \arg\max_{x^0,x^1,\ldots,x^L} P\left(y^0, y^1, \ldots, y^L \middle| x^0, x^1, \ldots, x^L\right) \\ &\quad \cdot P\left(x^0, x^1, \ldots, x^L\right) \\ &= \arg\min_{x^0,x^1,\ldots,x^L} \xi^{FF}\left(x^0, x^1, \ldots, x^L, y^0, y^1, \ldots, y^L\right) \\ &\quad + \xi^{LF}\left(x^0, x^1, \ldots, x^L\right). \end{aligned} \tag{6}$$

In the above equation, ξ^{FF} represents the energy of the feature field and ξ^{LF} represents the energy of the label field.

C. Relationship Among the Multigranularity Layers

A given remote sensing image I is segmented into K semantic classes using the MGOMRF-BC model, which requires an optimal solution of (6). The key lies in the probabilistic modeling problem of the hybrid feature field and the hybrid label field, i.e., $P(y^0, y^1, \ldots, y^L | x^0, x^1, \ldots, x^L)$ and $P(x^0, x^1, \ldots, x^L)$. However, it is very difficult to directly probabilistically construct them. As the two probability distribution functions contain a large number of random variables, it is difficult to give a specific joint functional probability form.

In the framework of the MGOMRF-BC model, there are $2L + 2$ modules used to simplify the two hybrid subrandom fields. The simplification is based on the following two assumptions.

Assumption 1: For any multigranularity layer, its feature field interacts only with: 1) its own label field and 2) the feature field and the label field of the previous layer in the transfer direction.

According to this assumption, the likelihood function of the hybrid feature field Y, $P(y^0, y^1, \ldots, y^L | x^0, x^1, \ldots, x^L)$, can be transformed into the form shown in the following equation:

$$\begin{aligned} &P\left(y^0, y^1, \ldots, y^L \middle| x^0, x^1, \ldots, x^L\right) \\ &\quad = P\left(y^0 \middle| x^0\right) \cdot \prod_{l=1}^{L} P\left(y^l \middle| x^l, y^{l-1}, x^{l-1}\right) \\ &\quad = P\left(y^L \middle| x^L\right) \cdot \prod_{l=0}^{L-1} P\left(y^l \middle| x^l, y^{l+1}, x^{l+1}\right). \end{aligned} \tag{7}$$

In traditional MRF-based methods, most of them are single-layer structures. Therefore, in feature field modeling, the

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assumption of independence between nodes' features is used for a given segmentation result. In MRF-based methods with multilayer structures, the assumption of independence between feature fields of each layer is mostly adopted. In this article, in order to explore the different granularities information of the target, we propose Assumption 1, which allows each granularity layer to capture the correlation between features at different granularities using feature information and semantic information of adjacent granularities. This is more convenient for the likelihood function to extract the features of the geographic targets more fully.

Assumption 2: For every multigranularity layer, its prior probability distribution of label field is only related to its own label field and the label field of the previous granularity layer in the transfer direction.

That is, the prior probability of the hybrid label field in (6), $P\left(x^0, x^1, \ldots, x^L\right)$, can be transformed into the form shown in the following equation:

$$\begin{aligned} P\left(x^0, x^1, \ldots, x^L\right) &= P\left(x^0\right) \cdot \prod_{l=1}^{L} P\left(x^l \middle| x^{l-1}\right) \\ &= P\left(x^L\right) \cdot \prod_{l=0}^{L-1} P\left(x^l \middle| x^{l+1}\right). \end{aligned} \tag{8}$$

For probabilistic modeling of the prior function of the hybrid label field, in this article, we propose Assumption 2, which can extend the original binary potential function defined only for a single layer to a binary potential function for a double layer. This enables probabilistic modeling of the label field to make more accurate probabilistic approximations based on the intra–inter layer semantic space distribution by semantic reasoning from different granularity perspectives.

With Assumptions 1 and 2, the two probabilities in (6) can be transformed into (7) and (8), respectively. Furthermore, it can be written as follows:

$$\begin{aligned} &P\left(y^0, y^1, \ldots, y^L \middle| x^0, x^1, \ldots, x^L\right) \cdot P\left(x^0, x^1, \ldots, x^L\right) \\ &= \left[P\left(y^0 \middle| x^0\right) \cdot P\left(x^0\right)\right] \cdot \prod_{l=1}^{L} \left[P\left(y^l \middle| x^l, z^{l-1}\right) \cdot P\left(x^l \middle| x^{l-1}\right)\right] \\ &= \left[P\left(y^L \middle| x^L\right) \cdot P\left(x^L\right)\right] \cdot \prod_{l=0}^{L-1} \left[P\left(y^l \middle| x^l, z^{l+1}\right) \cdot P\left(x^l \middle| x^{l+1}\right)\right]. \end{aligned} \tag{9}$$

In the above equation, $z^l = \left(x^l, y^l\right)$. Equation (9) is the probabilistic interpretation of the update path from coarse-grained to fine-grained and back to coarse-grained for L+1 granularity layers and is the theoretical basis for the green box part in Fig. 3. Specifically, the update path of the MGOMRF-BC model is given. First, the pixel granularity layer is updated independently to obtain the label results. After that, it is passed sequentially to the coarse granularity direction, and each layer needs to update the segmentation results according to the correlation of the above two assumptions at each pass until the Lth granularity layer. Then, the segmentation result of the Lth layer is updated independently. Finally, the updates are made sequentially in the fine-grained direction, and the correlation assumption is still used for each layer up to the pixel-granularity layer. Therefore, the update path for solving the segmentation results of the multigranularity structure can be viewed as a closed-loop path.



Fig. 4. Example of upper and lower neighborhoods. (a) Finer granularity layer over-segmentation results and neighborhoods. (b) Over-segmentation results and neighborhoods in this layer. (c) Coarser granularity layer over segmentation results and neighborhoods. (d) upper and lower neighborhoods of the corresponding region.

Since then, the originally complex probabilistic modeling of hybrid feature fields and hybrid label fields is now transformed into relatively simple probabilistic modeling based on the correlation between the granularity layers.

D. Probabilistic Modeling of Each Granularity

From (9), it can be seen that completing an update path requires two updates per granularity layer. The likelihood functions of the feature fields and the prior probabilities of the label fields for each granularity layer are modeled based on intra–inter-layer correlation. Thus, the neighborhood system at each granularity layer needs to be augmented in order to include the influence of adjacent granularity into the consideration of the probability modeling process. For the lth granularity layer, the neighborhood system at its own granularity can be given according to the RAG G^l, denote as N^l. The upper and lower neighborhoods of each layer are now defined as follows.

Definition (Upper and Lower Neighborhoods): For the lth granularity layer, the cite set of the region nodes in the $(l-1)$th granularity layer, which corresponds to S_i^l, is called the upper neighborhood of S_i^l and is denoted as $N_i^{l,l-1}$. The cite set of the region nodes in the $(l+1)$th granularity layer, which corresponds to S_i^l, is called the lower neighborhood of S_i^l and is denoted as $N_i^{l,l+1}$.

Fig. 4 is an illustration of the upper and lower neighborhoods. Fig. 4(a)–(c) shows the oversegmented results of the $(l-1)$th, lth, and $(l+1)$th granularity layers, respectively. Fig. 4(d) shows the set of intralayer, upper, and lower neighborhoods of the selected regions in Fig. 4(a)–(c). The sets of upper and lower neighborhoods for all nodes in the lth granularity layer are its upper and lower neighborhood systems; they can be denoted as $N^{l,l-1}$ (upper neighborhood system) and $N^{l,l+1}$ (lower neighborhood system). For the multigranularity structure, upper and lower neighborhood systems can be defined at granularity layers from 1 to $L-1$, while only the lower neighborhood system $N^{0,1}$ can be defined at the zeroth granularity layer, and only the upper neighborhood system $N^{L,L-1}$ can be defined at the Lth granularity layer.

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After introducing the definitions of upper and lower neighborhoods, (9) can be further written in the following form:

$$\begin{aligned}
&P\left(y^0, y^1, \ldots, y^L \middle| x^0, x^1, \ldots, x^L\right) \cdot P\left(x^0, x^1, \ldots, x^L\right) \\
&= \left[\prod_{i:i\in S^0} P\left(y_i^0 \middle| x_{N_i^0}^0\right) \cdot P\left(x_s^0 \middle| x_{N_s^0}^0\right)\right] \cdot \\
&\quad \cdot \left\{\prod_{l=1}^{L}\left[\prod_{i:i\in S^l} P\left(y_i^l \middle| x_{N_i^l}^l, z_{N_i^{l,l-1}}^{l-1}\right) \cdot P\left(x_i^l \middle| x_{N_i^l}^l, x_{N_i^{l,l-1}}^{l-1}\right)\right]\right\} \\
&= \left[\prod_{i:i\in S^L} P\left(y_i^L \middle| x_{N_i^L}^L\right) \cdot P\left(x_i^L \middle| x_{N_i^L}^L\right)\right] \\
&\quad \cdot \left\{\prod_{l=0}^{L-1}\left[\prod_{i:i\in S^l} P\left(y_i^l \middle| x_{N_i^l}^l, z_{N_i^{l,l+1}}^{l+1}\right) \cdot P\left(x_i^l \middle| x_{N_i^l}^l, x_{N_i^{l,l+1}}^{l+1}\right)\right]\right\}.
\end{aligned} \tag{10}$$

In the feature field modeling of multigranularity layers, it is known from the Assumption 1 and (10) that the modeling of the feature field likelihood function at each layer involves adjacent scales. To better express this correlation, the G-M model, which can describe the interactions between the features of each node, is used to model the feature field in this algorithm. Based on the results of the authors' previous work [60], [61], for a single-scale object-based G-M model, object-based linear regression equations can be performed separately for each region node, and Gaussian models can be modeled for the residual terms of the object-based linear regression equations, as detailed in [60] and [61]. The modeling is given as follows:

$$y_i^l - \mu_k^l = \sum_{i':i'\in N_i^l} \theta_{ii'}^l \cdot \left(y_{i'}^l - \mu_{x_{i'}^l}^l\right) + g_i^{l,k} \tag{11}$$

$$\theta_{ii'}^l = \frac{e_{ii'}^l}{e_i^l} \cdot \frac{\#s_{i'}^l}{\sum_{i':i'\in N_i^l} \#s_{i'}^l} \tag{12}$$

$$\begin{aligned}
P\left(y^l \middle| x^l\right) &= \prod_{i:i\in S^l} P\left(y_i^l \middle| x_i^l, z_{N_i^l}^l\right) \\
&= \prod_{i:i\in S^l} \frac{\exp\left[-\frac{1}{2} \cdot \left(g_i^{l,x_i^l}\right)^T \cdot \left(\Sigma_{x_i^l}^l\right)^{-1} \cdot g_i^{l,x_i^l}\right]}{\left(\sqrt{2\pi}\right)^p \cdot \left|\Sigma_{x_i^l}^l\right|^{\frac{1}{2}}}.
\end{aligned} \tag{13}$$

In the above equations, $\theta_{ii'}^l$ represents the feature interaction coefficient between regions s_i^l and $s_{i'}^l$; $\mu_{x_{i'}^l}^l$ is the feature vector mean of class $k = x_{i'}$ in layer l; $g_i^{l,k}$ is the residual term of the regression equation, which obeys Gaussian distribution; $g_i^{l,k} \sim N\left(0, \Sigma_k^l\right)$; $l = 0, L$; and Σ_k^l is the covariance matrix corresponding to the nodes of the kth semantic category.

For modeling the likelihood function of the feature field involving two granularity layers under the assumption of interlayer correlation, the object-level linear regression equation can be expanded from the upper and lower neighborhoods as follows:

$$\begin{aligned}
y_i^l - \mu_k^l = &\sum_{i':i'\in N_i^l} \theta_{ii'}^l \cdot \left(y_{i'}^l - \mu_{x_{i'}^l}^l\right) \\
&+ \sum_{i':i'\in N_i^{l,l+j}} \theta_{ii'}^{l,l+j} \cdot \left(y_{i'}^{l+j} - \mu_{x_{i'}^l}^{l+j}\right) + g_i^{l,k}.
\end{aligned} \tag{14}$$

In the above equation, $\theta_{ii'}^{l,l+j}$ denotes the proportion of the overlapping area of the region s_i^{l+j} in the $(l+j)$th layer and the region s_i^l in the lth layer to the area of s_i^l, calculated according to (15). $g_i^{l,k}$ is the regression residual of region s_i^l on semantic category k, which is a zero-mean Gaussian noise, i.e., $g_s^{l,k} \sim N\left(0, \Sigma_k^l\right)$, where Σ_k^l can be computed by the following equation:

$$\theta_{ii'}^{l,l+j} = \frac{\#\left\{(m,n)\middle|(m,n) \in s_i^l \cap s_{i'}^{l+j}\right\}}{\#\left\{(m,n)\middle|(m,n) \in s_i^l\right\}} \tag{15}$$

$$\Sigma_k^l = \frac{\sum_{i:x_i^l=k} \#\left\{(m,n)\middle|(m,n) \in s_i^k\right\} \cdot \left(g_i^k\right)^T \cdot g_i^k}{\sum_{i:x_i^l=k} \#\left\{(m,n)\middle|(m,n) \in s_i^k\right\}}. \tag{16}$$

The feature field likelihood function in (9) involving the dual-granularity layers can be written as follows:

$$\begin{aligned}
&P\left(y^l \middle| x^l, y^{l+j}, x^{l+j}\right) \\
&\quad = \prod_{i:i\in S^l} P\left(y_i^l \middle| x_i^l, z_{N_i^l}^l, z_{N_i^{l,l+j}}^{l+j}\right) \\
&\quad = \prod_{i:i\in S^l} \frac{\exp\left[-\frac{1}{2} \cdot \left(g_i^{l,x_i^l}\right)^T \cdot \left(\Sigma_{x_i^l}^l\right)^{-1} \cdot g_i^{l,x_i^l}\right]}{\left(\sqrt{2\pi}\right)^p \cdot \left|\Sigma_{x_i^l}^l\right|^{\frac{1}{2}}}.
\end{aligned} \tag{17}$$

With respect to the prior probability of the label field, pairewise potential is used to capture the interaction between different nodes. There are three types of interactions included for each node within each granularity layer due to the introduction of the interlayer-internal correlation assumption, as shown in Fig. 5. Specifically, the following holds.

1) The first type of interaction, which is the spatial representation of the semantic similarities and differences between s_i^l and its neighboring nodes $s_{i'}^l$ in the lth layer. It is denoted as $\xi_1^{LF}(x_i^l, x_{i'}^l)$.
2) The second type of interaction relationship, which is used to describe the change between the tth and $(t-1)$th segmentation labels of the s_i^l, is called intralayer blinking potential in this article. It is denoted as $\xi_2^{LF}(x_i^{l,t}, x_i^{l,t-1})$.
3) The third type of interaction relationship, which is used to describe the difference relationship between s_i^l and the $s_{i'}^{l+j} \in N_i^{l,l+j}$ of the $(l+j)$th granularity layer ($j = \pm 1$), is called the interlayer blinking potential in this article. It is denoted as $\xi_3^{LF}(x_i^{l,t}, x_{i'}^{l+j,t})$.

E. Self-Renewing Pairwise Potential With Blinking Coefficient

According to the update path of the MGOMRF-BC model and Assumption 2, the relationship between semantic labels of multigranularity layers can be modeled as a self-renewing pairwise potential with blinking coefficients. In the process of finding the optimal semantic segmentation of the multigranularity structure, this semantic inference model transforms the problem into the tth optimization of the lth granularity layer under the condition that the segmentations of the $(l+j)$th ($j = 1$ or -1) granularity layer are known

$$\begin{aligned}
&\hat{x}^{l,t}\left|x^{l,t-1}, y^l, x^{l+j,t}, y^{l+j}\right. \\
&\quad = \arg\max_{x^l} P\left(y^l \middle| x^l, z^{l+j,t}\right) \cdot P\left(x^l \middle| x^{l,t-1}, x^{l+j,t}\right)
\end{aligned}$$

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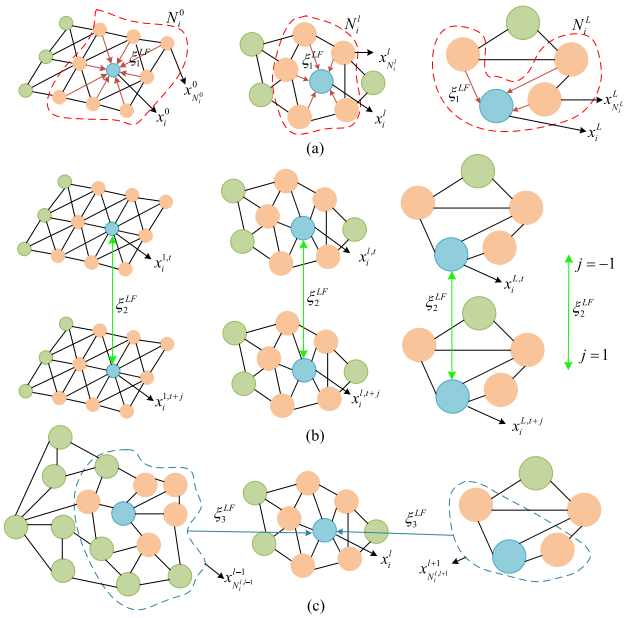


Fig. 5. Three components of the pairwise potential function: (a) intralayer neighborhood interactions, (b) historical information interactions at the same location, and (c) interlayer historical information interactions at the same location.

$$= \arg\min_{x^l} \xi^{FF}\left(y^l, x^l, z^{l+j,t}\right) + \xi^{LF}\left(x^l, x^{l,t-1}, x^{l+j,t}\right)$$
$$= \arg\min_{x^l}\Bigg\{\sum_{i:i\in S^l} \xi^{FF}\left(y_i^l, x_i^l, y_{N_i^l}^l, x_{N_i^l}^l, z_{N_i^{l,l+j}}^{l+j,t}\right) + \sum_{i:i\in S^l} \xi_1^{LF}\left(x_i^l, x_{N_i^l}^l\right) + \xi_2^{LF}\left(x_i^l, x_i^{l,t-1}\right) + \xi_3^{LF}\left(x_i^l, x_{N_i^{l,l+j}}^{l+j,t}\right)\Bigg\}. \tag{18}$$

The spatial semantic interaction with one granularity layer, $\xi_1^{LF}(x_i^l, x_{N_i^l}^l)$, is defined as

$$\xi_1^{LF}\left(x_i^{l,t}, x_j^{l,t-1}\right)_{i\in S^l, j\in N_i^l} = \begin{cases} -|i|\cdot\beta, & \text{if } x_i^{l,t} = x_j^{l,t-1} \\ |i|\cdot\beta, & \text{if } x_i^{l,t} \neq x_j^{l,t-1}. \end{cases} \tag{19}$$

Here, $|i|$ represents the number of pixels contained in i, which can introduce the granularity information into the potential function as a granularity-sensitive term. β is the potential parameter. The intralayer blinking potential term, $\xi_2^{LF}\left(x_i^{l,t}, x_i^{l,t-1}\right)$, is defined as

$$\xi_2^{LF}\left(x_i^{l,t}, x_i^{l,t-1}\right) = \begin{cases} -\left(1+\beta_{\text{intra}}^B\right)\cdot\left|\xi_2^{LF}\left(x_i^{l,t-1}, x_i^{l,t-2}\right)\right|, & \text{if } x_i^{l,t} = x_i^{l,t-1} \\ \left(1-\beta_{\text{intra}}^B\right)\cdot\left|\xi_2^{LF}\left(x_i^{l,t-1}, x_i^{l,t-2}\right)\right|, & \text{if } x_i^{l,t} \neq x_i^{l,t-1}. \end{cases} \tag{20}$$

β_{intra}^B is the intralayer blinking coefficient; this potential term forces the results of two updates of nodes at the same location to be as consistent as possible.

The interlayer blinking potential term, $\xi_3^{LF}(x_i^{l,t}, x_{N_i^{l,l+j}}^{l+j,t})$, is defined as

$$\xi_3^{LF}\left(x_i^{l,t} = k^l, x_{i'}^{l+j,t} = k^{l+j}\right)_{i':i'\in N_i^{l,l+j}} = -\beta_{\text{inter}}^{B.t}\cdot\left|\ln\left(\theta_{ii'}^{l,l+j}\cdot P_{k^l,k^{l+j}}^{l,l+j,t}\right)\right| \tag{21}$$

$$\beta_{\text{inter}}^{B.t} = \beta_{\text{inter}}^{B.t-1} + \chi_{x_{i'}^{l+j,t}=x_i^{l,t}}\cdot\frac{1}{T-t+1} \tag{22}$$

$$P_{k^l,k^{l+j}}^{l,l+j,t} = \frac{\left|\left\{(m,n)\middle|(m,n)\in i'\cap i, x_i^{l,t}=k^l, x_{i'}^{l+j,t}=k^{l+j}\right\}\right|}{\left|\left\{(m,n)\middle|(m,n)\in i', x_{i'}^{l+j,t}=k^{l+j}\right\}\right|}. \tag{23}$$

In the above equation, $\beta_{\text{inter}}^{B.t}$ is the interlayer blinking coefficient, $\theta_{ii'}^{l,l+j}$ is shown in (15), and $P_{k^l,k^{l+j}}^{l,l+j,t}$ is the label interlayer-transfer probability. This potential term can accelerate the label toward convergence.

Because $\xi_2^{LF}(x_i^{l,t}, x_i^{l,t-1})$ is the label of x_i^l in the $(t-1)$th cycle, it can be used as a priori information. Therefore, only the first iteration in the tth cycle of the lth granularity layer is taken into account, and the subsequent iterations in the tth cycle no longer consider $\xi_2^{LF}(x_i^{l,t}, x_i^{l,t-1})$. $\xi_3^{LF}(x_i^{l,t}, x_{N_i^{l,l+j}}^{l+j,t})$ is the result of the label of the adjacent granularity layer tth cycle, and each iteration in the tth solution process affects the lth granularity layer.

In most of the MRF-based semantic segmentation algorithms, the termination conditions are set as the number of cycles does not exceed the preset value, or the difference between the results obtained from the two adjacent cycles is 0. In this article, the termination condition is set as no difference between the results obtained from the two adjacent cycles, or the number of cycles reaches 10 based on the reference of most MRF-type algorithms and the actual performance of the proposed algorithm. The experiments of setting the number of cycles are described in the experimental section.

The algorithm of the MGOMRF-BC model is given in Algorithm 1. In order to generate layers with different granularity, we use the more mature and open code oversegmentation algorithms, mean shift (MS), and simple linear iterative clustering (SLIC). l_1 and l_2 layers are generated, respectively.

IV. Experiment Results

In order to verify the effectiveness of the proposed MGOMRF-BC algorithm for the segmentation of semantic ambiguous regions in remote sensing images, this section will be tested on different remote sensing image datasets. First, the adopted dataset and the evaluation index of segmentation results will be briefly introduced. After that, the corresponding setting strategies for the manual input parameters involved in the proposed algorithm will be given through experiments. Finally, a comparison with other MRF-based algorithms will be made. All MRF-based algorithms are run on HP Laptop15-dc1xxx (CPU: Intel[1] Core[2] i7-9750H CPU @ 2.60 GHz; GPU: GTX1660Ti; RAM: 32-GB 2666-MHz DDR4; and

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Algorithm 1 Algorithm of the MGOMRF-BC Model

Input: remote sensing image I; number of semantic categories K; potential function parameter β; blinking coefficients β^B_{intra}, $\beta^{B,0}_{\text{inter}}$; MS-generated granularity layer l_1 and SLIC-generated granularity layer l_2

Output: the segmentation result $\hat{x}$

Based on I, a multi-granularity layer structure with $L = l_1 + l_2 + 1$ layers is formed by MS algorithm and SLIC algorithm.
Get the site set of nodes at each granularity layer $S = \{S^l | 0 \le l \le L\}$, and the region adjacency graph (RAG) $G = \{G^l | 0 \le l \le L\}$
Use the OMRF algorithm to obtain the initial semantic segmentation results for each layer $x^0 = \{x^{l,0} | 0 \le l \le L\}$.
$t = 0$, $x^{-1} = 0$, $diff_x = \sum_{l=0}^{L} |x^{l,1} - x^{l,0}|$.
while sum$(diff_x(:)) > 0$ and $t \le T$ **do**
  Let $t = t + 1$, $z^{l,t-1} = (y^l, x^{l,t-1})$.
  Get segmentation result of 0-th layer, $x^{0,t}$.
  for $l = 1 : 1 : L$ **do**
    Based on Eq.18, obtain $x^{l,t}$.
  end for
  Let $t = t + 1$, $z^{l,t-1} = (y^l, x^{l,t-1})$.
  Get segmentation result of L-th layer, $x^{L,t+1}$.
  for $l = L - 1 : -1 : 0$ **do**
    Based on Eq.18, obtain $x^{l,t}$.
  end for
  $diff_x = \sum_{l=0}^{L} |x^{l,t} - x^{l,t-1}| + |x^{l,t-1} - x^{l,t-2}|$, $x^{t-1} = x^t$, Let $t = t$ - 1.
end while
Let $\hat{x} = x^{l_1+1,t}$, return $\hat{x}$.

ROM: 512 GB), operating system is Windows 10 Professional 64-bit; system version is 19043.1526; and the software is MATLAB2020a. All deep-learning algorithms are run on Dell Precision 3660 (CPU: Intel[1] Core[2] i7-12700 K CPU @ 3.60 GHz; GPU: RTX3090; RAM: 32-GB 4400-MHz DDR5; and ROM: 1 TB); operating system is Windows 11 Professional 64-bit; and system version is 22621.1702.

A. Datasets and Evaluation Metrics

In this experiments section, nine images from the synthetic texture image dataset [62], the SPOT-5 dataset, and the Gaofen Image Dataset (GID) are selected. Among them are three synthetic texture images, including two synthetic texture images based on GeoEye remote sensing images and one synthetic color texture image; the spatial resolution of SPOT-5 images is 10 m, and the dimensions of the three images are 369 × 369, 256 × 256, and 1300 × 1300, which contains four semantic categories: vegetation, urban area, farmland, and water. The space resolution of images selected from GID [63] is 1 m, and the dimensions of the three images are 2400 × 2000, 2500 × 2500, and 2600 × 3000, which contains three semantic categories: water, farmland, and urban area.

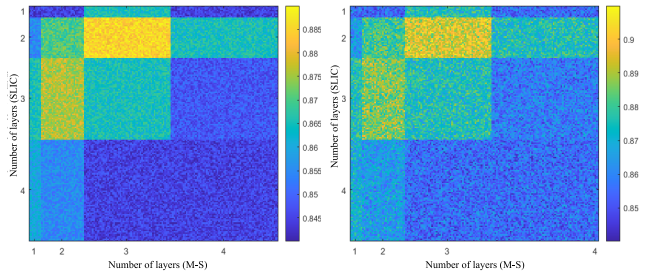


Fig. 6. Kappa and OA for different numbers of granularity layers (l_1, l_2) and regional granularity (mra, nor): (a) Kappa for different settings and (b) OA for different settings.

In order to measure the performance of various algorithms for semantic segmentation of remote sensing images in terms of segmentation accuracy, intersection over union (IoU), mean IoU (m-IoU), Kappa, and overall accuracy (OA) are selected as evaluation metrics of segmentation accuracy in this section.

B. Test of MGOMRF-BC Model

The MGOMRF-BC algorithm involves three main types of parameters: 1) the number of granularity layers L (l_1 layers generated by MS and l_2 layers generated by SLIC) and the region granularity parameters (the minimum region area (mra) in the MS algorithm and the number of oversegmented regions (nor) in the SLIC algorithm); 2) the potential energy function parameters (β, β^B_{intra}, and β^B_{inter}) in the MRF model for each granularity layer; and 3) the next experiments are conducted to explore the setting strategies for each of these parameters.

1) Number of Granularity Layers and Regional Granularity Settings: The settings of the number of granularity layers and region granularity are directly linked to the complexity of the overall algorithm. Different numbers of granularity layers and granularity sizes can perform different oversegmentations for the same geophysics target to extract different granularity features. To better balance the computational speed and accuracy of the results, this section takes values 1, 2, 3, and 4 for l_1 and l_2, respectively, to test the segmentation performance of the algorithm. In order to test out the appropriate region granularity of the algorithm, the mra for the layer generated by MS is set to 0.02%, 0.04%, 0.06%, 0.08%, 0.10%, 0.12%, 0.14%, and 0.16% of the image area, respectively. The nor for the layers generated by SLIC is set to 5000, 5500, 6000, 6500, 7000, 7500, 8000, and 8500, respectively. Specifically, the l_1 layer generated by MS and the l_2 layer generated by SLIC can take $C_8^{l_1} \times C_8^{l_2}$ groups of different area granularity combinations. A randomly selected image from the test image is used for the experiment. A GF-2 image (GF-2 image 2, the second image for GID data comparison experiments) is selected for this experiment, and it contains three semantics: urban area, farmland, and water. In order that the experiments are not affected by other parameters, the other parameters are set to the same values for each group of experiments. Fig. 6(a) and (b) shows the color grid of Kappa and OA for each group of experiments, respectively.

From Fig. 6, we can see that the segmentation accuracy of the algorithm is higher when $l_1 = 3$ and $l_2 = 2$. This is because, when the number of layers is set too large, the

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overall computational complexity of the algorithm rises, and the segmentation results may be oversmoothed; while the number of layers, if set too low, cannot form an effective granularity distinction, and the oversegmentation results of each granularity layer are basically the same, it is impossible to give the distinction of each layer, which is not conducive to the blinking region convergence. Therefore, in the subsequent experiments, the strategy for setting the granularity layers of MGOMRF-BC algorithm is $l_1 = 3$ (the number of MS layers) and $l_2 = 2$ (the number of SLIC layers).

2) Potential Energy Function Parameters: The potential energy function parameter is a very important parameter in the MRF model, which is mainly used to balance the role of the feature field and the label field for the semantic segmentation of each node. In order to obtain the range of setting interval of potential function parameters in the MGOMRF-BC model and the relationship of the potential function parameter between each granularity layer, the following exploratory experiments are conducted in this section. The potential function parameter setting strategies for the remaining granularity layers are, respectively, given as follows:

$$\beta^{*,l} = k \cdot \beta^{*,0} + b, \quad \beta^* \in \left\{\beta, \beta^B_{\text{intra}}, \beta^B_{\text{inter}}\right\} \tag{24}$$

$$\beta^{*,l} = \log_{l+1}\beta^{*,0}, \quad \beta^* \in \left\{\beta, \beta^B_{\text{intra}}, \beta^B_{\text{inter}}\right\} \tag{25}$$

$$\beta^{*,l} = 0.5^l \cdot \beta^{*,0}, \quad \beta^* \in \left\{\beta, \beta^B_{\text{intra}}, \beta^B_{\text{inter}}\right\}. \tag{26}$$

For the pixel-granularity layer l_0, β^0,$\beta^{B,0}_{\text{intra}}$, and $\beta^{B,0}_{\text{inter}}$ take values from 0.1 to 4 in steps of 0.1. When testing the potential energy function parameters, the rest of the parameters are kept constant, and Kappa and OA are used as measures. The results for the three types of parameters in the self-renewing two-point potential energy function are shown in Fig. 7. That is, first, keep β^B_{intra} and β^B_{inter} constant and test the effect of the value of β on the accuracy of the algorithm, as shown in Fig. 7(a) and (d). After that, the best setting strategy and interval of β are chosen to be fixed, while β^B_{inter} is fixed to test the segmentation accuracy when β^B_{intra} is taken to different setting strategies and values, as shown in Fig. 7(b) and (e). Finally, β and β^B_{intra} are taken the appropriate values and setting strategies at the same time, and the interval and setting strategy of β^B_{inter} is tested, as shown in Fig. 7(c) and (f). As can be seen from Fig. 7(a) and (d), when the value of β^0 exceeds 1.3, the accuracy of the corresponding segmentation result starts to decrease sharply. This is because, when β^0 is too large, the label field has too large effect on the segmentation results, and the semantics of the large regions in the segmentation results will continuously swallow the small regions, thus making the segmentation results less accurate. Only when β^0 takes values in the interval of 0.7–1.3, the segmentation results of all three potential function parameter setting strategies are in the stable interval, among which the segmentation results with the highest accuracy are set according to the strategy of (24). As for the intralayer blinking coefficient and the interlayer blinking coefficient, the overall change trend is similar to that of β, but the fluctuation of the change in segmentation accuracy is smoothed out in a certain interval. This is because the inclusion of the intralayer blinking coefficient and the interlayer blinking coefficient makes the model take the historical label information of the corresponding positions within and between layers into account, which can appropriately accelerate the regional convergence during the update cycle and, at the same time, can keep the semantic labels from changing repeatedly. According to the results in Fig. 7, the strategy for setting the three types of potential function parameters in the rest of the experiments in this article is that the parameters of the pixel-level granularity layer take values in the interval of 0.7–1.3, and the potential function parameters of the remaining granularity layers are set according to the strategy of (24).

3) Number of Cycles: To verify that the proposed MGOMRF-BC segmentation algorithm can converge within a small number of cycles and has a faster convergence rate than other MRF-based methods, this section takes an SPOT-5 image (SPOT-5 image 2, the second image for SPOT-5 data comparison experiments) as an example, which contains three categories: urban area, water, and farmland. In order to more intuitively represent the convergence of each algorithm, this section uses the energy E^t_{method} and the energy reduction rate $\Delta E^t_{\text{method}}$ of the segmentation results obtained by each algorithm after each cycle as a measure. This is calculated as follows:

$$\begin{aligned} E^t_{\text{method}}(I) &= E^t_{\text{method}}(Y|X) + E^t_{\text{method}}(X) \\ &= -\sum_{(m,n)\in S}\left[\ln P^t\left(Y_{m,n}\middle|X_{m,n}\right) + \ln P^t\left(X_{m,n}\right)\right] \end{aligned} \tag{27}$$

$$\Delta E^t_{\text{method}} = \frac{E^{t+1}_{\text{method}} - E^t_{\text{method}}}{E^t_{\text{method}}}. \tag{28}$$

N_{method} in (27) denotes the number of nodes in the method, and t is the number of cycles. Fig. 8(a) and (b) shows the energy and the energy reduction rate of the MGOMRF-BC model with the other six methods in each update cycle.

As shown in Fig. 8, the initial cycle energy of all OMRF-based algorithms is close to the final convergence energy of the pMRF algorithm. This is because these object-based algorithms use the final results of the pixel-based algorithm as the a priori segmentation for the initial cycle of segmentation label optimization. The energy of each segmentation result of the MRF-based algorithm decreases as the number of cycles increases. The MGOMRF-BC algorithm proposed in this article has a faster energy drop compared to other methods. This is because the comparison method updates the segmentation tokens in each cycle based only on the semantic performance of the surrounding neighborhood of the current node, while the method proposed in this article considers not only the surrounding neighborhood nodes in this layer but also the last semantic result of the current node in this layer and the semantics of the nodes in the corresponding positions of the adjacent granularity layers when updating in each granularity layer, and this processing optimization method speeds up the iterative convergence speed. Moreover, the MGOMRF-BC algorithm has less variation in the total energy of segmentation results after eight cycles, and the segmentation results tend to be smooth. Therefore, for insurance reasons, the number of cycle updates of the MGOMRF-BC algorithm is set to 10 in all other experiments in this section.

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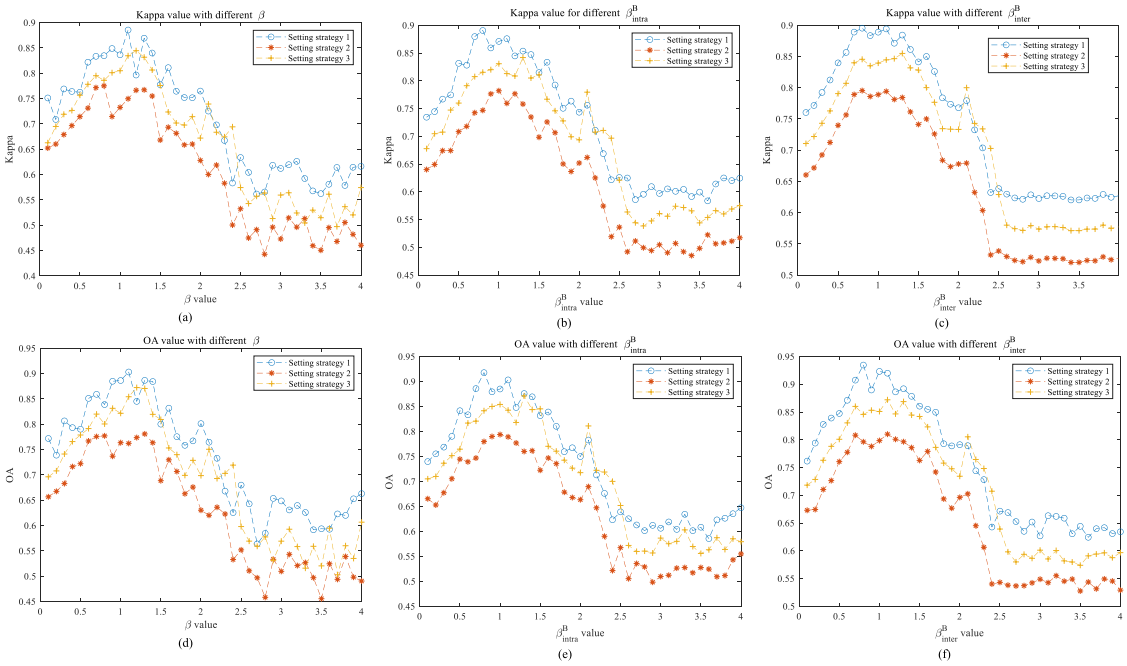


Fig. 7. Experiment of setting the strategy of parameters in the potential energy function. (a) Kappa values of the experimental results with different values of different setting strategies for β. (b) Kappa values of the experimental results with different values of different setting strategies for β^B_{intra}. (c) Kappa values of the experimental results with different values of different setting strategies for β^B_{inter}. (d) OA values of the experimental results with different values of different setting strategies for β. (e)OA values of the experimental results with different values of different setting strategies for β^B_{intra}. (f) OA values of the experimental results with different values of different setting strategies for β^B_{inter}.

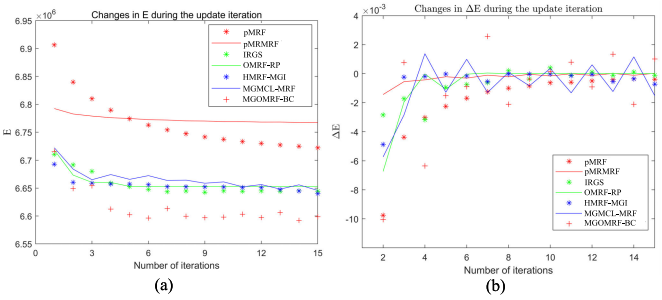


Fig. 8. Experiment of setting the number of cycles. (a) Energy of the segmentation result after each cycle. (b) Energy reduction rate of the segmentation result after each cycle.

Fig. 9 shows the experiment about the convergence of the algorithm proposed in this article, and the experimental image is selected from the SPOT5 remote sensing images. Among them, Fig. 9(a) represents that the algorithm is updated sequentially with each big update containing six granularity layers when it performs a cyclic update; Fig. 9(b) shows the segmentation results obtained from the initialization step, the first cycle, and the last cycle of the algorithm update process. As can be seen from Fig. 9(a) and (b), the energy of the segmentation results is continuously decreasing as the algorithm performs the update, and there is a large energy reduction in the initialization step and the intermediate steps. This is due to the fact that the initial segmentation of each object-granularity layer is obtained with one update of the pixel-level results. The energy reduction that occurs in the intermediate process is due to the algorithm stepping out of the local steady state by considering the spatiotemporal correlation and avoiding the local optimal results.

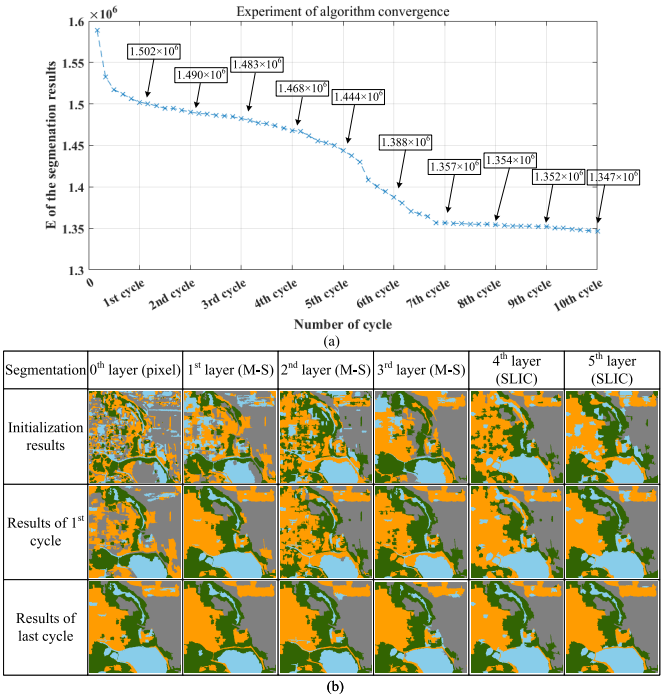


Fig. 9. Experiments on algorithm convergence. (a) Energy of the segmentation result after each big cycle. (b) Segmentation results of initialization, first, and last cycles.

C. Comparison Experiments

In order to test the semantic segmentation of the MGOMRF-BC algorithm in remote sensing images, this section will set up a comparison experiment between the MGOMRF-BC algorithm and other MRF-based algorithms.

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The experiments are tested on four remote sensing image datasets: synthetic texture image synthesized by the Prague texture generator, SPOT-5 remote sensing image, Gaofen-2 remote sensing image, and near-ground aerial remote sensing image. The comparison methods include the following six main types.

1) *pMRF [64]:* The classical pixel-based MRF model, where the nodes in the algorithm are pixel points, is solved in an ICM model.
2) *pMRMRF [50]:* A pixel-based multiresolution MRF model with a multilayer structure, each with a different resolution.
3) *IRGS [65]:* It optimizes the OMRF model by considering the edge strength information and estimating the model parameters in a more stable manner.
4) *OMRF-RP [29]:* The object-based MRF method with the single-layer structure and unique region granularity, introducing the idea of anisotropy penalty in the label field to enhance the recognition of small targets.
5) *HMRF-MGI [55]:* Introduce the idea of multigranularity to consider the effects of both pixel-level granularity and region-level granularity in the probabilistic modeling of label fields.
6) *MGMCL-MRF [46]:* Introduce the idea of multigranularity to consider the effects of both pixel-level granularity and region-level granularity in the probabilistic modeling of label fields.
7) *DeepLabv3+ [66]:* A deep learning algorithm with encoder–decoder architecture, capturing multiscale features and fusing them with the help of ASPP module to obtain higher accuracy segmentation results.
8) *SegFormer [67]:* A deep learning semantic segmentation algorithm based on transformer, which uses a hierarchical structure of transformer encoder to output multiscale features.
9) *BiSeNet V2 [68]:* An efficient bilateral semantic partitioning network architecture where two paths are used to handle spatial details and category semantics, respectively, and a new aggregation layer are designed to enhance the interconnection between the two paths.

The comparison between the MGOMRF-BC model and different types of MRF-based models can be obtained by comparing with the above models. The pMRF is a single granularity method, pMRMRF is a multilayer method, IRGS and OMRF-RP are single granularity methods with full consideration of regional factors in the label field, OMRF-A is a single granularity with multisemantic method, and HMRF-MGI and MGMCL-MRF are multigranularity methods. DeepLabv3+ is a deep learning algorithm based on an encoding-decoding structure that can extract multiscale information. SegFormer is a transformer-based method; BiSeNet V2 is a multibranch network structure. All three of them are classical deep learning algorithms in the field of semantic segmentation. To ensure the fairness of the comparison experiments, the potential function parameters of all methods refer to the setting strategy in Section IV-B. The oversegmentation algorithms in the object-based models with a single region granularity all use the MS algorithm with an mra, 0.09% of the image; other parameters refer to the optimal interval of each method.

TABLE I

QUANTITATIVE INDEXES OF SEVEN MRF-BASED METHODS FOR SEMANTIC SEGMENTATION OF TEXTURE SYNTHETIC IMAGES

Method	Fig.10		Fig.11		Fig.12	
	Kappa	OA	Kappa	OA	Kappa	OA
pMRF	59.69	64.98	69.81	74.49	45.12	49.88
pMRMRF	67.21	71.88	81.79	85.27	67.38	7436
IRGS	84.73	88.41	80.95	85.02	64.28	69.10
OMRF-RP	87.64	90.87	82.32	86.11	64.38	69.20
HMRF-MGI	88.53	91.59	84.30	87.87	66.07	70.88
MGMCL-MRF	91.96	93.82	90.47	92.71	89.17	91.66
MGOMRF-BC	**97.96**	**98.52**	**96.50**	**97.32**	**91.35**	**93.33**

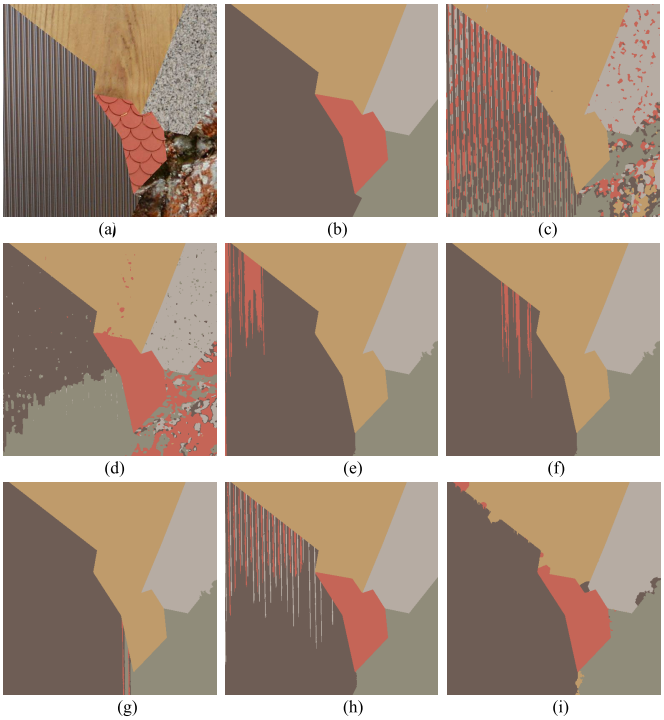


Fig. 10. Segmentation results of the normal color synthetic texture image. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of HMRF-MGI. (h) Result of MGMCL-MRF. (i) Result of MGOMRF-BC.

1) Texture Synthetic Images: First, experiments are conducted on the texture synthetic images obtained from Prague texture generator, as shown in Figs. 10(a)–12(a), which are a normal natural color synthetic texture image and two synthetic texture images stitched by GeoEye remote sensing images, both of size 512 × 512, where the spatial resolution of GeoEye remote sensing synthetic texture images is 0.5 m. Due to the small sample of synthetic texture images, deep learning algorithms are not used for comparison. The segmentation results of each comparison algorithm are shown in Figs. 10–12, and the quantitative evaluation indexes of each segmentation result, Kappa coefficient, and OA coefficient are shown in Table I.

For the normal color texture composite image, as shown in Fig. 10(a), the lower left part is a region consisting of striped color bands. Due to the obvious differences in color features in this part, pMRMRF cannot group them together. The pMRMRF captures a larger range of neighborhood space through a multiresolution structure, but this region still cannot

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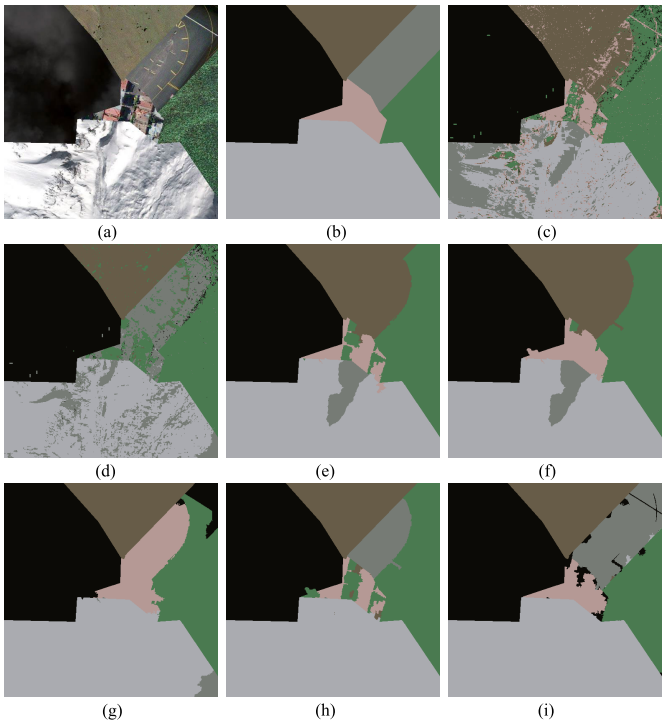


Fig. 11. Segmentation results of the GeoEye remote sensing synthetic texture image. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of HMRF-MGI. (h) Result of MGMCL-MRF. (i) Result of MGOMRF-BC.

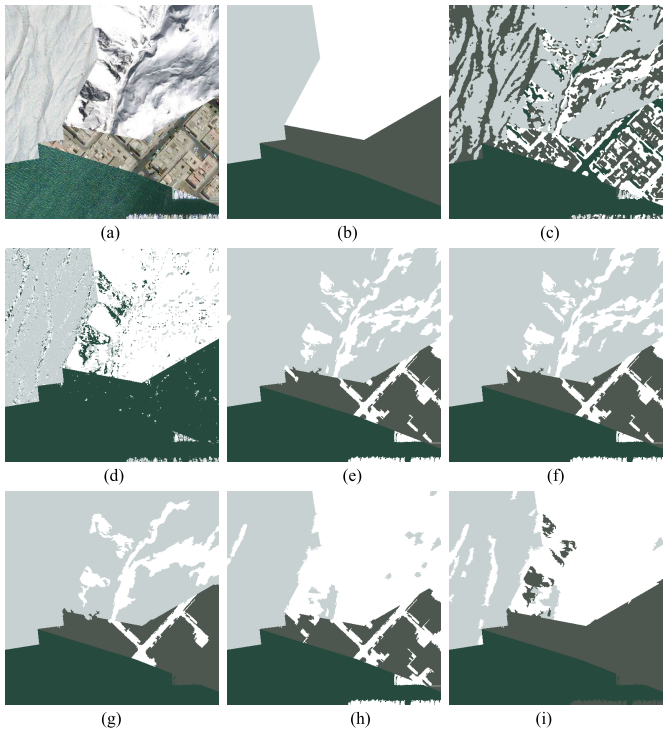


Fig. 12. Segmentation results of another GeoEye remote sensing synthetic texture image. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of HMRF-MGI. (h) Result of MGMCL-MRF. (i) Result of MGOMRF-BC.

be grouped into one category. Several other OMRF-based methods use the pMRF results as the initial segmentation, so there are some misclassified pixels in this part as well. It can also be seen that pMRF, IRGS, OMRF-RP, and HMRF-MGI all classify the red part and the wood grain part in the middle of the image as one category, but pMRMRF, MGMCL-MRF, and MGOMRF-BC can segment these two types of textures. This is because, with the multilayer structure, the classification results can be verified separately at each layer and will not be trapped in local optimization because there is only one layer. From Table I, it can be seen that the segmentation results of MGOMRF-BC are the best. The MGOMRF-BC method uses a multiscale structure to obtain oversegmentation results on the original image, which minimizes the probability of unreasonable region segmentation labels due to inaccurate initial oversegmentation results. At the same time, in each cycle of processing, not only the object-based G-M model is constructed on each granularity layer but also the segmentation results of each granularity layer are compared before and after, and between layers to form a more reasonable intra–inter-layer information decision transfer process. Therefore, the algorithm proposed in this article can achieve the segmentation task of color synthetic texture images with higher accuracy than the selected contrast algorithm.

For the GeoEye remote sensing texture synthesis image, the segmentation results are shown in Figs. 11 and 12. The corresponding Kappa coefficients and OA coefficients are shown in Table I. The upper right part of Fig. 11(a) is always misclassified as other category segmentation results. For Fig. 11(a), pMRMRF, HMRF-MGI, MGMCL-MRF, and MGOMRF-BC have a multilayer structure, and the results of each layer can be updated and verified separately, which reduces the phenomenon of repeated changes in the category of similar regions and, thus, improves the accuracy of segmentation results. Among them, the segmentation accuracy of the MGOMRF-BC method is higher. This is because, in the case of similar RGB features, pixel-level segmentation methods cannot capture region-level features due to the small range of neighborhoods considered, and the parts with similar features are always classified into one class. For object-level algorithms, oversegmented regions can be segmented as nodes, and higher accuracy can be achieved by considering a larger neighborhood range. However, there are also many misclassifications. This is because, when an oversegmented region is obtained, some neighboring pixels of different classes with similar features may be divided into one region, or small differences between pixel-level features are smoothed when computing region-level features. During segmentation, some oversegmented regions of different categories that are adjacent and have similar RGB features may be misclassified as the same category. The MGOMRF-BC method uses different oversegmentation algorithms to oversegment the original image to form a multigranularity scale region structure. In the oversegmentation stage, it minimizes the possibility of oversegmented regions containing pixels of different classes. In addition, during the iterative process, the object-based G-M model first updates the segmentation labels at each scale layer, and then, regions with different segmentation labels between each scale layer will further calculate the labeled field potential

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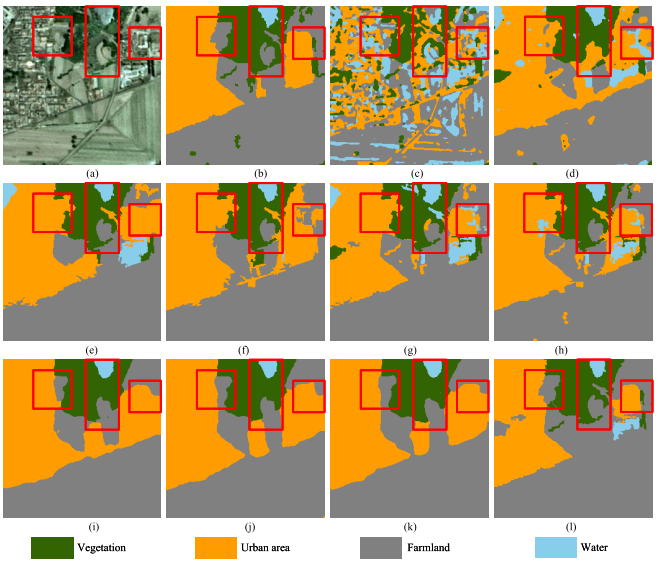


Fig. 13. Segmentation results of SPOT-5 image 1. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of OMRF-A. (h) Result of MGMCL-MRF. (i) Result of DeepLabv3+. (j) Result of SegFormer. (k) Result of BiSeNet V2. (l) Result of MGOMRF-BC.

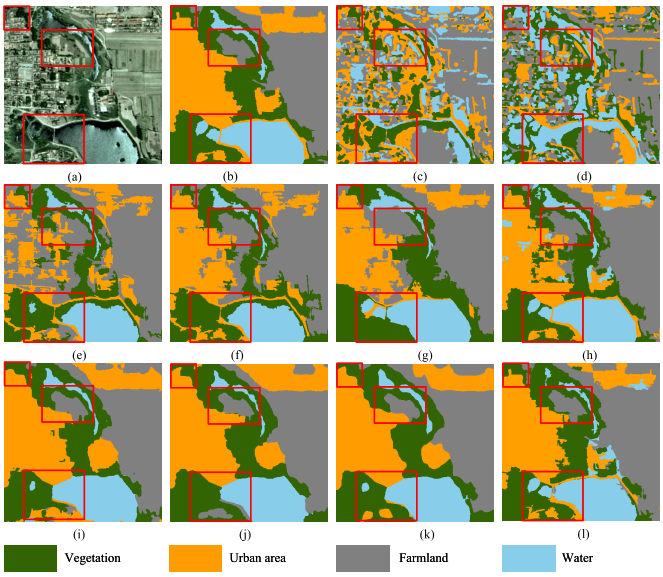


Fig. 14. Segmentation results of SPOT-5 image 2. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of OMRF-A. (h) Result of MGMCL-MRF. (i) Result of DeepLabv3+. (j) Result of SegFormer. (k) Result of BiSeNet V2. (l) Result of MGOMRF-BC.

function by means of an intra–inter-layer potential function with blinking coefficients.

For Fig. 12(a), the two types of feature textures in the upper half are very close in RGB feature distribution, which directly causes the segmentation results of pMRF, IRGS, OMRF-RP, and HMRF-MGI to have large misclassification in these two categories. The pMRF segmentation results show a large number of sporadic misclassified areas of pepper salts. pMRMRF misclassifies the lower half of the two feature textures into the same category, but the division is relatively complete, especially in the water. Segmentation results of IRGS, OMRF-RP, and HMRF-MGI have most of the upper right area misclassified into the same category as the upper left area, but a few of them are correctly classified. However, a small number of them are correctly classified because the RGB of this part is clearly different from the RGB of other similar feature areas. MGMCL-MRF basically distinguishes the upper part of the image due to the introduction of multiple semantic auxiliary layers, but MGMCL-MRF still fails to classify the lower right region into one category, mainly because of the large differences between streets and residential buildings, and the Gaussian mixture model using the independence assumption for modeling the feature field cannot fully utilize the features of the corresponding locations at different granularities of other auxiliary layers. MGOMRF-BC, on the other hand, achieves a unified degree of differentiation between the upper part of the image and the lower right region through the feature expression and semantic expression correlation role of the feature targets at different granularities.

2) SPOT-5 Remote Sensing Image: In this section, three SPOT-5 real remote sensing images of Pingshuo are used as experimental objects, as shown in Figs. 13(a), 14(a), and 15(a), all with a spatial resolution of 10 m. All three images contain four categories: arable land, water bodies, residential areas,

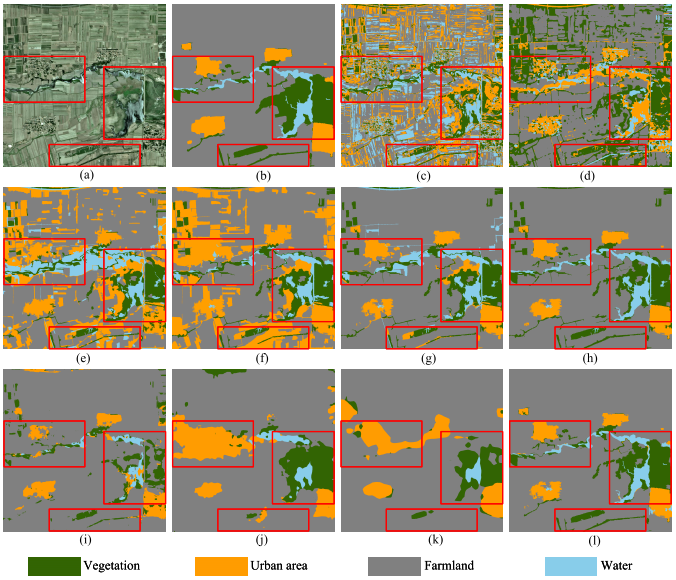


Fig. 15. Segmentation results of SPOT-5 image 3. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of OMRF-A. (h) Result of MGMCL-MRF. (i) Result of DeepLabv3+. (j) Result of SegFormer. (k) Result of BiSeNet V2. (l) Result of MGOMRF-BC.

and vegetation. The image size of Fig. 13(a) is 256 × 256, that of Fig. 14(a) is 369 × 369, and that of Fig. 15(a) is 1300 × 1300. The segmentation results of several compared methods are shown in Figs. 13–15, and the quantitative evaluation indexes of segmentation results of each algorithm, Kappa coefficient, and OA coefficient are shown in Table II.

To test the semantic segmentation performance of the deep learning category algorithm on SPOT5 images, we provided 1024 training sample images sized 100 × 100 with precise semantic results to DeepLabv3+. As for SegFormer and BiSeNet V2, we use 472 images of different sizes (including

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TABLE II
QUANTITATIVE INDEXES OF SEVEN MRF-BASED METHODS AND THREE DEEP-LEARNING-BASED METHODS FOR SEMANTIC SEGMENTATION OF SPOT-5 IMAGES

Test Image	Indexes (%)	pMRF	pMRMRF	IRGS	OMRF-RP	HMRF-MGI	MGMCL-MRF	DeepLabv3+	SegFormer	BiSeNet V2	MGOMRF-BC
	Vegetation	53.70	72.07	75.48	73.16	70.44	77.19	**83.59**	74.31	74.22	82.76
	Urban area	33.65	64.24	70.27	75.22	76.21	76.65	80.36	82.62	84.33	**85.70**
	Farmland	56.37	78.54	84.54	82.74	**92.83**	90.53	74.14	86.08	87.29	90.25
Fig.13	Water	2.85	19.69	23.30	80.22	21.23	29.78	85.07	**85.73**	84.36	38.58
	m-IoU	36.64	58.63	63.40	77.83	65.17	68.54	80.79	82.18	**82.55**	74.32
	Kappa	51.65	75.61	80.42	82.60	84.11	85.20	84.68	86.28	87.19	**88.83**
	OA	59.45	82.05	86.35	88.10	89.47	90.10	89.59	90.84	91.55	**92.90**
	Vegetation	37.88	29.20	64.20	72.93	58.67	82.78	84.60	79.68	78.37	**85.19**
	Urban area	15.67	19.62	52.29	66.25	67.58	80.49	80.33	79.37	81.79	**86.48**
	Farmland	39.31	39.95	74.22	70.91	71.08	86.77	75.54	79.53	84.76	**94.25**
Fig.14	Water	10.01	12.08	43.13	66.12	**90.61**	85.07	81.42	82.69	83.86	81.09
	m-IoU	25.72	25.21	58.46	69.05	71.98	83.78	80.47	80.32	82.20	**86.75**
	Kappa	36.66	37.37	65.95	76.98	77.70	88.61	86.18	85.86	87.31	**91.37**
	OA	42.10	42.46	71.02	81.19	81.70	91.11	89.14	88.86	90.07	**93.40**
	Vegetation	26.45	22.30	46.40	49.07	49.03	61.50	**85.93**	49.62	35.66	80.64
	Urban area	5.70	29.19	21.34	20.10	70.46	**88.71**	56.24	27.14	42.92	87.17
	Farmland	52.20	67.80	67.30	69.24	91.17	94.33	30.63	81.80	85.04	**96.72**
Fig.15	Water	8.39	8.24	19.34	45.60	31.17	74.13	44.71	43.05	22.47	**87.27**
	m-IoU	23.19	31.88	38.60	46.00	60.46	79.67	54.38	50.40	46.52	**87.95**
	Kappa	38.94	51.41	54.49	56.68	75.55	85.41	67.47	64.86	65.51	**91.50**
	OA	50.92	66.75	68.59	70.70	89.38	94.28	85.76	81.88	84.15	**96.95**
Average Kappa		42.42	54.80	66.95	72.09	79.12	86.41	79.44	79.00	80.00	**90.57**
Average OA		50.82	63.75	75.32	80.00	86.85	91.83	88.16	87.19	88.59	**94.42**

254 images sized 200 × 200, 148 images sized 350 × 350, and 70 images 900 × 900) as training samples. The semantic segmentation results of these deep learning algorithms are shown in (i)–(k) in Figs. 11–13.

It can be seen from Table II that the MGOMRF-BC method has the best segmentation accuracy for Figs. 13(a), 14(a), and 15(a). MGOMRF-BC has significantly better segmentation results than other comparison algorithms for the regions marked by red boxes in the images. In the segmentation results of Figs. 13(a), 14(a), and 15(a), yellow represents urban areas, blue represents water, gray represents farmland, and green represents vegetation.

For Figs. 13(a) and 14(a), the pixel-level segmentation method and the pMRF method have a large number of misegmentations and present fragmented segmentation results. This is because the pixel-level MRF method cannot consider a larger range of neighborhood regions, so only the four or eight neighborhoods of the current point can be modeled in the update iteration, and the segmentation results have poor regional consistency. The pMRMRF can consider macroscopic and microscopic neighborhoods by constructing a multiscale resolution structure, so the segmentation results are improved over pMRF, but it is still a pixel-based iterative update model, which inevitably has a large area of misclassification. Object-level MRF segmentation methods generally have higher accuracy but still differ. IRGS only considers pixels at the edges of adjacent regions of the current oversegmented region and does not use other region-level information of adjacent neighborhoods, so the segmentation accuracy is not very high. During the update iteration of OMRF-RP, not only the region-level color space features of the oversegmented region itself but also some region-level spatial information of the oversegmented region, such as region area and region boundary length, are taken into account to improve the segmentation accuracy. The HMRF-MGI method integrates pixel-level influencing factors and area, and influencing factors in the label field, and fully considers area consistency and edge accuracy, so the segmentation performance is also better but still classifies the area as vegetation class in the urban area without considering the neighborhood environment of the residential area in which it is located. MGMCL-MRF uses different semantic levels and different granularities to construct four sub-MRF modules and uses the segmentation results of the adjacent module to further improve the final segmentation accuracy. However, due to the introduction of multisegmentation auxiliary layers, some regions in the auxiliary layers cannot be classified accurately, resulting in information transfer mechanisms that affect the classification inaccuracy of the final results. From the segmentation indicators, for Figs. 13(a) and 14(a), DeepLabv3+ has higher accuracy for recognizing vegetation categories in images, but there is region oversmoothing, and the recognition of urban area and farmland edges is poor. The SegFormer algorithm has higher accuracy for the recognition of water but poor recognition of the edges between farmland and vegetation, causing some farmland to be misclassified as vegetation. The segmentation accuracy of BiSeNet V2 is relatively high for farmland, and the overall recognition accuracy (m-IoU, Kappa, and OA) is relatively high on small-size images. In the upper right and middle regions of Fig. 13(a), there are missegmentation regions among urban areas, farmland, and vegetation. In the lower left region and the upper left region of Fig. 14(a), there are missegmentation areas between urban area and vegetation; there are also some missegmentation areas among water, urban area, and vegetation. The MGOMRF-BC method constructs a multigranularity structure in the initial oversegmentation region process, establishes the G-M model based on interlayer correlation, respectively, and then considers the influence of the same granularity layer neighborhood on the current node segmentation label in the label field modeling while also integrating the blinking information of the previous update label in the corresponding position of the same layer and the neighborhood layer to accelerate the convergence speed of segmentation label update as much as possible.

For Fig. 15(a), the farmland in the original image has a small area of vegetation showing the RGB feature distribution known from the vegetation class semantic area, a small amount of low shrub area in the vegetation area is similar to the farmland in terms of RGB feature distribution, a small amount

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TABLE III
QUANTITATIVE INDEXES OF SEVEN MRF-BASED METHODS AND THREE DEEP-LEARNING-BASED METHODS FOR SEMANTIC SEGMENTATION OF GF-2 IMAGES

Test Image	Indexes (%)	pMRF	pMRMRF	IRGS	OMRF-RP	HMRF-MGI	MGMCL-MRF	DeepLabv+	segformer	BiSeNet V2	MGOMRF-BC
Fig.16	Urban area	42.39	60.55	54.00	53.05	54.03	51.06	**72.76**	69.55	65.18	53.93
	Farmland	51.87	47.76	64.76	65.75	65.05	71.99	86.13	**82.23**	78.62	**71.90**
	Water	21.76	21.73	40.77	41.21	**42.24**	56.43	68.99	**91.51**	88.63	81.07
	m-IoU	38.67	43.35	53.18	53.34	53.77	59.83	75.96	**81.10**	77.47	68.97
	Kappa	50.59	53.72	63.45	63.58	63.83	67.43	**81.96**	80.94	**77.71**	70.09
	OA	59.53	61.59	73.79	74.22	74.13	78.60	**89.12**	87.90	85.33	**79.91**
Fig.17	Urban area	38.17	40.62	49.65	49.44	80.60	88.78	92.89	84.49	92.35	**94.02**
	Farmland	55.60	87.43	90.90	90.08	70.74	**93.41**	85.53	70.27	86.34	**88.64**
	Water	15.34	13.24	15.06	15.34	**89.99**	44.13	76.36	71.17	84.89	**87.13**
	m-IoU	36.37	47.09	51.87	51.62	80.44	75.44	84.93	75.31	87.86	**89.93**
	Kappa	49.02	53.01	58.37	58.58	**79.93**	86.17	89.77	79.87	90.22	**92.06**
	OA	56.00	60.18	66.14	66.36	87.27	91.93	94.56	88.67	94.81	**95.84**
Fig.18	Urban area	45.05	63.59	65.89	69.68	75.56	**97.06**	42.12	37.56	51.62	**94.64**
	Farmland	68.05	67.11	90.29	92.78	91.45	96.24	72.35	65.56	80.91	**97.21**
	Water	12.17	16.12	**86.11**	82.73	82.90	57.80	**72.83**	75.26	78.69	56.84
	m-IoU	41.76	48.97	80.76	81.73	83.30	**83.70**	62.43	59.46	70.41	82.89
	Kappa	52.74	59.08	82.97	85.37	85.84	91.92	65.21	60.90	72.76	**92.25**
	OA	68.04	71.78	91.59	93.13	93.35	96.86	77.05	72.00	84.02	**97.00**
Average Kappa		50.78	55.27	68.26	69.18	76.53	81.84	78.98	73.90	80.23	**84.80**
Average OA		61.19	64.52	77.17	77.90	84.92	89.13	86.91	82.86	88.05	**90.92**

of tree aggregation area inside the urban area is visually close to the vegetation, and the water contains very few pixels. All these make semantic segmentation more difficult. In the segmentation results of pMRF and pMRMRF, a large area of misclassification occurs in the farmland, and a large number of pixels in the urban area are classified as vegetation in the segmentation results of pMRF. The segmentation results of IRGS and OMRF-RP misclassify a large number of farmlands as urban areas. This is mainly because the RGB distribution of the two is closer. The segmentation results of HMRF-MGI further improve the phenomenon that it is difficult to distinguish urban areas from farmland due to the alternate update of multigranularity, and urban areas are basically identified correctly, while a small amount of farmland is misclassified as vegetation. Low shrub areas in vegetation were misclassified as farmland in the segmentation results of HMRF-MGI, MGMCL-MRF, and MGOMRF-BC. The segmentation results of MGMCL-MRF basically complete the identification of farmland and urban areas, and it introduces multisemantic auxiliary label fields, so that it can maintain high segmentation accuracy on large-size remote sensing images with large intraclass differences and small interclass differences. MGOMRF-BC, on the other hand, introduces the assumption of interlayer feature correlation and a new label field pairwise potential function in a multilayer structure, which can better transfer and retain segmentation information at each scale in the multilayer structure. The segmentation results of DeepLabv3+, in terms of indicators, and the segmentation accuracy of all three semantic categories are not high except for the vegetation category. From the segmentation images, there are large misclassifications between water and urban areas, and it also happens in the segmentation between farmland and vegetation. The segmentation results of SegFormer, from the indicators, only the farmland part reach more than 80%, and the IoU indices of the other three categories do not exceed 50%. From the semantic segmentation images, there are large misclassifications, especially the confusion between the urban area and the farmland, and the water part is also mostly misclassified into the vegetation and the urban area category. The segmentation results of BiSeNet V2 show an oversmoothing phenomenon; and large water regions are missegmented into an urban area and farmland; and the edges of each semantic category region are very inaccurate. All the above phenomena can be summarized as image refinement segmentation defects caused by deep learning algorithms that cannot effectively use image spatial information because of convolutional and pooling operations.

3) GF-2 Remote Sensing Image: In this part, three GF-2 remote sensing images selected from the GID database are used as the experimental objects, as shown in Figs. 16(a)–18(a). The spatial resolution of the full-color images of GF-2 remote sensing satellite is 1 m, and the sizes of the selected images are 2400 × 2000, 2500 × 2500, and 2600 × 3000. All three selected images contain three types of feature semantics: urban area, farmland, and water. The segmentation results produced by each comparison method are shown in Figs. 16–18, and the accuracy evaluation indexes, the Kappa coefficient, and the OA coefficient of the segmentation results are shown in Table III.

In the segmentation results of Figs. 16–18, the blue area is the water, the green area is the farmland, the red area is the urban area, and the black area in the real segmentation map is the area in the dataset where no real label is given. The RGB features of residential areas in these three selected high-definition remote sensing images are similar to the RGB features of water body areas, so the segmentation results of pMRF and pMRMRF, two MRF methods that only consider a small range of pixel neighborhoods, are poor and present a large area of misclassification.

To test the semantic segmentation performance of the deep learning category algorithm on GF-2 images, we selected 140 images of size 7200 × 6800 with precise semantic results from the GID database to train the semantic segmentation capability of the deep learning algorithms for GF-2 remote sensing images. The semantic segmentation results of these deep learning algorithms are shown in (i)–(k) in Figs. 16–18. MGOMRF-BC has significantly better segmentation results than other comparison algorithms for the regions marked by black boxes in the images.

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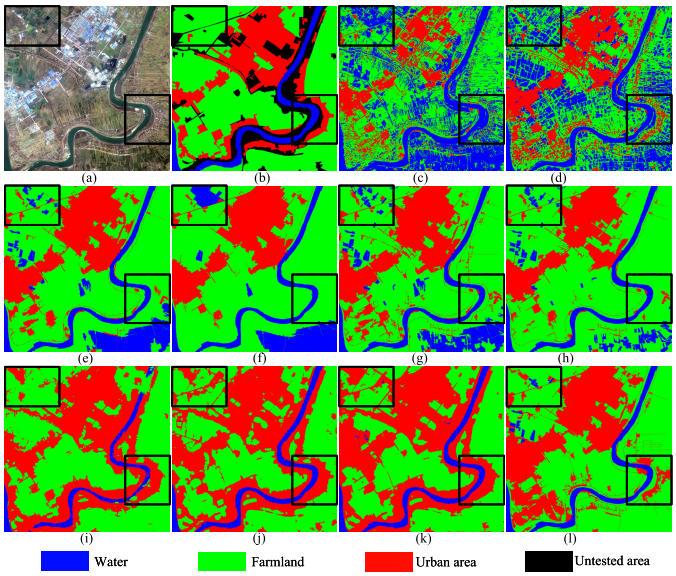


Fig. 16. Segmentation results of GF-2 image 1. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of OMRF-A. (h) Result of MGMCL-MRF. (i) Result of DeepLabv3+. (j) Result of SegFormer. (k) Result of BiSeNet V2. (l) Result of MGOMRF-BC.

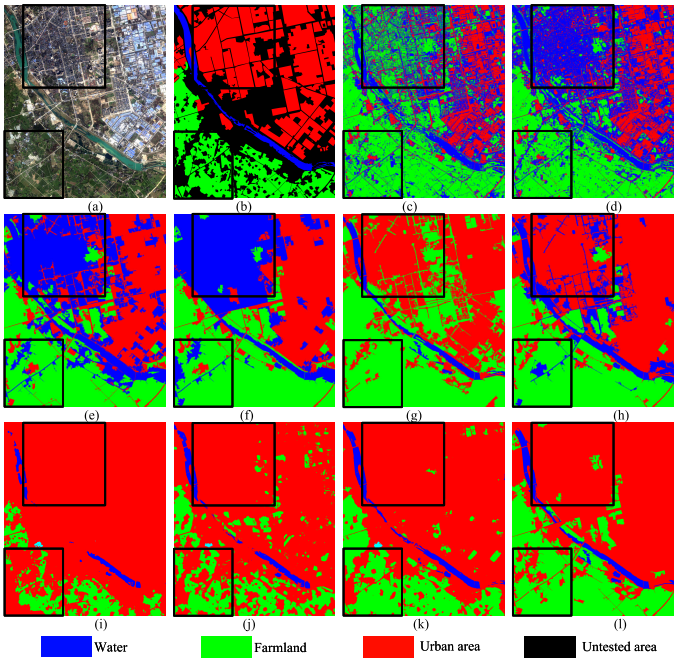


Fig. 17. Segmentation results of GF-2 image 2. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of HMRF-MGI. (h) Result of MGMCL-MRF. (i) Result of DeepLabv3+. (j) Result of SegFormer. (k) Result of BiSeNet V2. (l) Result of MGOMRF-BC.

For Fig. 16(a), the RGB features of the farmland in the lower right region show green, which is far from the RGB features of other farmland regions, but close to the water. The distribution of the urban area features around the water is similar to that of the farmland. Therefore, in the object-level processing method using the single-peaked Gaussian distribution model for feature field modeling, the segmentation results of IRGS, OMRF-RP, HMRF-MGI, and MGMCL-MRF all have large areas of green farmland misclassified as water in the lower right corner. In contrast, the MGOMRF-BC

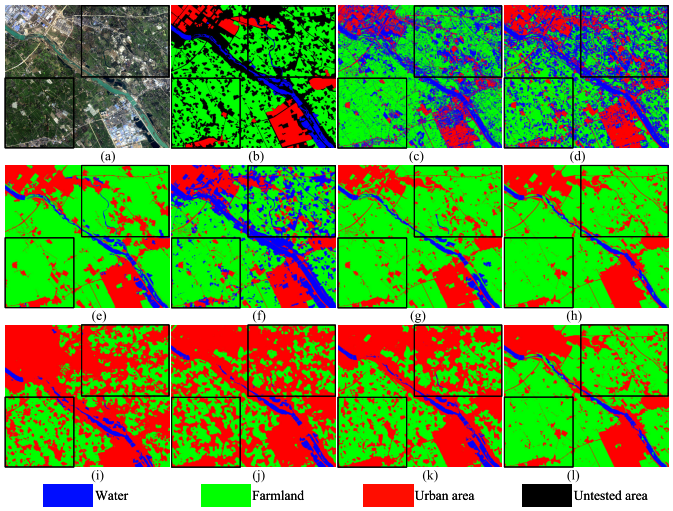


Fig. 18. Segmentation results of GF-2 image 3. (a) Original image. (b) Ground truth. (c) Result of pMRF. (d) Result of pMRMRF. (e) Result of IRGS. (f) Result of OMRF-RP. (g) Result of HMRF-MGI. (h) Result of MGMCL-MRF. (i) Result of DeepLabv3+. (j) Result of SegFormer. (k) Result of BiSeNet V2. (l) Result of MGOMRF-BC.

algorithm with feature field modeling using the object-level G-M model with the assumption of upper and lower layers granularity feature correlation performs well in the lower right corner segmentation. However, for the deep learning algorithms, their segmentation results are significantly better than the MRF-based algorithms. For urban area segmentation, DeepLabv3+ has the highest accuracy, while, for the other three categories, the model with the highest recognition accuracy is SegFormer. BiSeNet V2 also has a significant improvement compared to MRF-based algorithms. However, from the segmentation results, the deep learning algorithm identifies both sides of the river as urban areas, while the MRF-based algorithm is closer to the real situation for both sides of the river. The inaccurate labeling of the segmentation truth map is also one of the main reasons for the low segmentation evaluation index of the MRF-based algorithm.

In the segmentation task of Fig. 17(a), the RGB features of the left and right parts of the urban area category are obviously different, so the algorithms that use the single-peaked Gaussian model as the feature field modeling approach do not perform well in the segmentation of the urban area category. While the HMRF-MGI introduces multigranularity information into the model, and the MGMCL-MRF introduces the auxiliary semantics and multigranularity information, which improve the segmentation result, the segmentation results of DeepLabv3+ and SegFormer have a lot of missegmentation results, which divides the large area of water and farmland into the urban area. The segmentation results of BiSeNet V2 have higher accuracy in the segmentation of farmland and urban area. However, it can be seen from the real image that the area on both sides of the river is mainly a staggered distribution of urban area and farmland rather than a single urban area, but, since the annotation of the semantic category of the true value map does not give accurate annotation results for this part, the deep learning algorithm performs well in terms of IoU indicators for each category alone. From the actual image distribution, the deep learning algorithm has

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poor segmentation results for both sides of the river, and the recognition of urban areas in the farmland area is also very rough. MGOMRF-BC further improves the segmentation accuracy of blinking regions in remote sensing images by modeling the feature fields with object-level G-M models through the correlation assumption of each scale layer and by fully considering the spatiotemporal neighborhoods under the multiscale structure when modeling the label fields in each granularity layer of the multilayer structure.

For Fig. 18(a), there are a large number of blue roofs in the urban area, which are more different from other subfeature targets in the urban area and are easily misclassified into other categories in the segmentation process. The water itself accounts for a smaller proportion, and the distribution of RGB features is close to that of the farmland. All these bring challenges to the segmentation process. The segmentation results of DeepLabv3+ and SegFormer show a large range of urban area and farmland missegmentation, mainly in the lower left and upper right areas of the image. The missegmentation results of BiSeNet V2 have improved in the lower left area, but there are still a large number of missegmentation areas in the upper right area. These phenomena also indicate that the deep learning algorithm mainly focuses on the extraction of deep features, and a large number of convolutional pooling operations causes the edges of each small region in the segmentation results to be too smooth, which also reflects the insufficient utilization of spatial information. In the segmentation results of pMRF and pMRMRF, a large number of blue-roofed urban areas are misclassified into water regions, while water regions are misclassified into farmland regions. This is mainly because the pixel-level method can only consider a smaller spatial extent during the label update, and the label field is not as useful as the feature field, so the segmentation results with close RGB features are all classified into one category. The object-level method is improved, and water regions are basically segmented and identified in the result of IRGS, but a large number of urban areas are still misclassified as water. OMRF-RP and HMRF-MGI, on the other hand, misclassify a large number of urban areas as farmland. This is mainly because the urban areas in the lower right corner of the image are actually undeveloped plant wastelands, which are closer to farmland in terms of RGB distribution. The segmentation results of both MGMCL-MRF and MGOMRF-BC are relatively satisfactory. Both introduce multigranularity information, MGOMRF-BC is an object-level G-M model based on the assumption of interlayer correlation in feature field modeling, and the label field introduces a blinking potential function, which more fully exploits the interlayer information at different granularity layers.

4) Results' Discussion: From the above comparison experiments, it can be seen that the proposed algorithm in this article performs better in the semantic segmentation task of remote sensing images. In particular, for the parts boxed in black in each comparison experimental figure, it can be seen that the algorithm proposed in this article performs better for the preservation of region edges and segmentation of fine-grained objects. This is because, compared with other selected MRF methods, the proposed method for the use of multigranularity information is more consistent with the characteristics of the image spectral feature distribution. Also, the proposed method can effectively establish the interaction between different granularities, and the potential energy function with blinking coefficients in the spatiotemporal neighborhood is used, which also provides a more accurate description of the semantic space. Compared with the deep learning algorithms, the proposed method in this article effectively avoids the phenomenon of oversmoothing for region boundaries. This is because deep learning algorithms mainly use convolution and pooling operations for deep feature extraction, but this will cause problems such as blurred region boundaries and the inability to make full use of spatial information, which will eventually cause misclassification of segmentation results for some regions with fuzzy semantics. The proposed algorithm adopts a multigranularity approach for feature extraction, the association between feature expressions at different granularities for the same geographic target is also taken into account in the modeling, and the semantic spatial information and historical temporal information of this layer and adjacent granularity layers are fully considered in the process of segmentation marker division so that the recognition of region boundaries is more accurate.

D. Experimental Analysis

In Sections III-B–III-E, the multigranularity structure modeling, the intra–inter-layer feature correlation modeling in the feature field, and the spatiotemporal-neighborhood-based potential energy function with blinking coefficient in the labeled fields are introduced in detail. In the above experiments, the setting strategies for the parameters in MGOMRF-BC and the termination conditions of the algorithm are given. The proposed algorithm is compared with some MRF-based algorithms and deep learning algorithms. These experiments verified that MGOMRF-BC can improve the segmentation effect for semantically ambiguous small regions as much as possible while ensuring the overall segmentation accuracy of the image. This is mainly due to the use of multigranularity information and the full consideration of spatiotemporal neighborhood information in the updating process, which avoids the repeated transformation of the semantics of regions with fuzzy semantics. To further analyze the proposed algorithm, ablation experiments are conducted in this section for the innovative and improved parts of the algorithm, respectively.

MGOMRF-BC mainly makes innovations and improvements in three aspects: 1) the construction of multigranularity structure; 2) the use of the G-M model based on the assumption of intra–inter-layer correlation for the feature field likelihood function; and 3) the use of the spatiotemporal neighborhood-based model with blinking coefficients for the label field potential energy function. Both points 2 and 3 are based on the multigranularity structure; therefore, the ablation experiments are designed as follows. The multigranularity G-M model is constructed with points 1 and 2 as method A; the multigranularity spatiotemporal blinking model is constructed with points 1 and 3 as method B; and the MGOMRF-BC is constructed with all points included. The

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TABLE IV
EVALUATION INDICATORS FOR ABLATION EXPERIMENTS ON SPOT5 AND GF-2 IMAGES

		Fig.13	Fig.14	Fig.15	Fig.16	Fig.17	Fig.18
Method A	mIoU	68.21	81.93	80.83	58.47	80.97	80.62
	Kappa	84.98	87.90	86.34	66.92	84.86	89.37
	OA	90.04	90.87	94.96	75.89	89.70	94.98
Method B	mIoU	68.45	82.95	81.94	59.53	81.89	82.51
	Kappa	85.03	88.14	87.15	67.29	85.94	91.31
	OA	90.06	90.84	95.26	78.31	90.86	96.17
MGOMRF-BC	mIoU	**74.32**	**86.75**	**87.95**	**68.97**	**89.93**	**82.89**
	Kappa	**88.83**	**91.37**	**91.50**	**70.09**	**92.06**	**92.25**
	OA	**92.90**	**93.40**	**96.95**	**79.91**	**95.84**	**97.00**

evaluation indicators of the ablation experiments are shown in Table IV.

Combining Tables II–IV, it can be seen that both Method A and Method B have a large improvement compared to the pixel-level algorithm, IRGS, and HMRF-MGI. HMRF-MGI is an improved MRF algorithm involving only multigrain size structure, so both Method A and Method B are improved compared with HMRF-MGI. However, compared to MGMCL-MRF, the improvement is not much, basically the same. This is because MGMCL-MRF has added an auxiliary marker field to increase the accuracy of semantic segmentation, as in the case of multigrain structures. Comparing Method A and Method B, Method B has higher segmentation accuracy, which indicates that the potential function with flicker coefficients based on the spatiotemporal neighborhood has a better improvement effect on semantic segmentation than the interlayer-internal correlation GM model.

V. CONCLUSION

In this article, a new MGOMRF-BC model is proposed to integrate multigranularity information, and the semantic inference mechanism based on spatial–temporal neighborhoods is proposed. The strength of the proposed model mainly lies in two aspects. One aspect is the MGOMRF-BC framework, which introduces the multigranularity layers structure, and the specific intra–inter-layer feature association representation model is added to the MRF-based approach. Another aspect is the self-renewing pairwise potential function with blinking coefficients, which can effectively utilize the spatial neighborhood information and history information of semantic labels under multigranularity layers structure to accelerate the convergence of regions with repeatedly transformed semantic categories. The performance of the algorithm depends on the settings of each parameter, and the setting strategies of each parameter involved in the algorithm are given through exploratory experiments. Meanwhile, by conducting experimental tests on different remote sensing image datasets, it is verified that the proposed algorithm in this article performs better in segmenting the semantic segmentation problem compared with other MRF algorithms based on multiple granularity or labeled field improvement.

At present, the MGOMRF-BC model proposed in this article only uses two oversegmentation algorithms to generate a multigranularity layers structure, but it can be extended to using more oversegmentation algorithms to generate different granularity layers with different oversegmentation rules so that the model can further explore the deeper granularity information in the image. In other words, MGOMRF-BC can have a "deeper layer" structure for extracting granularity features and constructing pairwise potential energy functions.

Due to the strategy of updating paths, the complexity of the current algorithm is high. Our next research will focus on how to simplify the model and obtain a lightweight structure.

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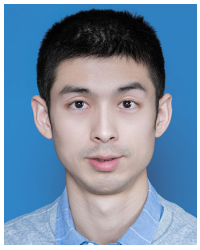
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